CEBU INSTITUTE OF TECHNOLOGY – UNIVERSITY

College of Computer Studies

Bachelor of Science in Information Technology

Software Test Document (STD v1.0)

PlayIT

An Offline-First Gamified Early Literacy Mobile Application

Course	IT411 (Section G7) — Capstone & Research 2 | Semester 1, AY 2026–2027
Document Version	1.0.1 (test record of debug APK H, commit `0320385`)
Publication Date	October 18, 2026
Document Status	Submitted for Adviser Review and Evaluation
Prepared by	Group 56 — PlayIT Capstone Team: Riva, Zendrix (Team Lead); Palis, John James; Miel, Kaysean; Durano, Ashlee Skye; Bien, Erik Samuel
Adviser	Mr. Joemarie C. Amparo

Table of Contents

Document Revision History 4
1. Introduction 4
1.1 Purpose 4
1.2 Test Item and Scope 4
1.3 References 5
1.4 Definitions 5
2. Test Approach 5
2.1 Test Levels 5
2.2 Test Design Techniques 6
2.3 Beyond the Happy Path 6
2.4 Pass/Fail, Entry and Exit Criteria 7
2.5 Test Environment 7
3. Traceability: SRS → Features → Test Cases → Results 7
4. Test Case Specifications and Results 8
4.1 Module 1: Letter-Sound Acquisition (Hear It) 9
4.2 Module 2: Speech Production and Tutoring (Say It) 10
4.3 Module 3: Phoneme Discrimination (Find It) 12
4.4 Module 4: Word Construction (Blend It) 14
4.5 Module 5: Progression, Gamification and Profiles 16
4.6 Module 6: Parent Zone 18
4.7 Cross-Cutting Requirements 19
4.8 Test Cases Not Run on This Build 22
5. Test Execution 23
5.1 Execution Record 23
5.2 Results by Test Class 23
5.3 Skipped Tests 26
5.4 Inspections 26
5.5 Device Tests on the Samsung Galaxy A21s 26
6. Layout Verification 27
6.1 Method 27
6.2 Results of the Layout Checks 27
6.3 Visual Review of the Captures 29
7. Incident and Defect Report 33
7.1 Incidents Found and Closed 34

7.2 Open Defects and Observations 35
8. Test Summary Report 35
8.1 Results in Numbers 35
8.2 Assessment 36
8.3 Next Steps 36
8.4 Sign-Off 36
Appendix A. Test Execution Log 36
Appendix B. How to Reproduce 50

Document Revision History

Version	Date	Primary Author(s)	Description
1.0	October 14, 2026	Capstone Team	First issue. Specifies the test cases for SRS v3.0 rev. 3.8, records a clean run of the whole automated suite on the code of debug APK H (433 test cases: 427 passed, 6 skipped by design, 0 failed), the 4-size layout matrix, two inspections of the build, and the incidents found in testing. It reports two open defects found while writing it (DEF-01, DEF-02). The tables of results are generated from the raw JUnit logs by `tools/docs/build_std.py`, which stops if a cited test is missing or did not pass.
1.0.1	October 18, 2026	Capstone Team	The test item is now the team lead's build of the same commit `0320385` (SHA-256 `b5a54c18…42d4`, 108,317,310 bytes), the file installed on the team's Samsung Galaxy A21s and submitted for the midterm. The build inspected on October 14 (`d301007f…937d`, 108,489,471 bytes) was made from that commit on another PC. The two differ only in the debug signing key, the line endings of 8 text files of the speech model, and code that the compiler and Hilt generate; no class written by the team, picture or sound differs (`docs/std/evidence/2026-10-18-submitted-apk/apk_inspection.txt`). The submitted file was inspected on October 18 with the same permissions as on October 14 (§5.4). No automated test result changed. The team then ran the acceptance check of the submitted file on the A21s (§5.5, October 18: pass), and the team lead reviewed and signed this version (§8.4).

1. Introduction

1.1 Purpose

This Software Test Document records how the refactored PlayIT was tested against its requirements, and what the tests showed. It follows the chain the course asks for:

Updated SRS (v3.0 rev. 3.8) → System Features (the six modules) → Test Cases (§4) → Test Execution (§5, §6) → Test Results (§5 to §8)

For every test case it states the input, the expected result taken from the SRS, the actual result observed, and the evidence (the test methods that ran, or the inspection output). A test case is marked **Pass** only when every piece of its evidence passed in the recorded run. Where the build does not meet a requirement, the test case is marked **Fail** and the defect is reported in §7, not left out. Where a test case could not run on this build (a planned feature, a field measure, or a device timing), it is listed in §4.8 with the reason and the date it is scheduled.

The structure follows ISO/IEC/IEEE 29119-3:2021 (test documentation), condensed for one build: test plan (§2), test case specifications (§4), execution log (§5, Appendix A), incident report (§7) and test completion report (§8).

1.2 Test Item and Scope

- **Test item:** PlayIT debug APK H, `playit-debug-H-0320385.apk` (SHA-256 `b5a54c18…42d4`), built from branch `refactor/hear-say-it` at commit `0320385` on the team lead's PC. This is the build submitted with the SDD v2.0 rev. 2.8, SRS v3.0 rev. 3.8 and SPMP v2.0 rev. 2.6. An earlier build of the same commit (SHA-256 `d301007f…937d`) was inspected on October 14; revision 1.0.1 above records the change.
- **In scope:** all automated unit, component and layout tests in `app/src/test/`; the 4-size layout matrix; an inspection of the APK's permissions; an integrity check of every released audio clip; the phone

tests the team ran on earlier builds, which found the incidents in §7.1; and the acceptance check of the submitted APK on the phone (§5.5).

- **Out of scope for this run** (each listed in §4.8): Round 2 field validation with learners and teachers (Weeks 7–8), the Gate 3 teacher audit of audio, timing measured on the device, and the features the SDD §2.0 lists as planned (telemetry and CSV export, spaced recall, session pacing, the voice-driven mic ripple).

1.3 References

1. PlayIT Software Requirements Specification (SRS v3.0, rev. 3.9), October 18, 2026: requirement IDs and the RTM test case IDs (§4).
2. PlayIT Software Design Description (SDD v2.0, rev. 2.9), October 18, 2026: §2.0 implementation status, §6 verification plan.
3. PlayIT Software Project Management Plan (SPMP v2.0, rev. 2.6): WBS 7 (Verification and STD), Milestones 6 and 7.
4. PlayIT MVP Validation Findings and Refactoring Priorities (course form), October 18, 2026.
5. ISO/IEC/IEEE 29119-3:2021, *Software and systems engineering — Software testing — Part 3: Test documentation*.
6. IEEE Std 829-2008, *IEEE Standard for Software and System Test Documentation* (superseded by 29119-3; used for the incident report fields).

1.4 Definitions

Term	Meaning in this document
Test case (TC)	One check of one requirement behaviour, with input, expected result, actual result and status. IDs follow the SRS RTM families (TC-AUD, TC-MIC, TC-ASR, TC-BLD, TC-ACC, TC-IND, TC-DB, TC-TEL, TC-EXP, TC-REV, TC-PERF, TC-SCR) and add TC-SAY, TC-FND, TC-PRG, TC-PZ, TC-SEC and TC-POL for behaviours the RTM groups together.
Test method	One JUnit test in `app/src/test/`; a test case cites one or more.
Pass / Fail / Skip	As recorded by JUnit. A skip is allowed only for the 6 font-scale checks on the 360x640 profile (§5.3).
Not run	The test case is specified but could not run on this build; §4.8 gives the reason.
Robolectric	Runs Android code, including Jetpack Compose screens, on the JVM, so screens can be tested without a phone.
Roborazzi	Captures the screens rendered by Robolectric as images (§6).
Unclipped bounds	A component's full position, even if it is scrolled or pushed out of view; the layout checks use it so that content off screen fails.
Foil	A wrong answer that sounds close to the right one: the letter's name ("em" for /m/), an added vowel ("muh"), or a substituted sound ("pa" for /f/).
dp / sp	Density-independent pixels (size) and scaled pixels (text).

2. Test Approach

2.1 Test Levels

Level	What it exercises	How it runs	Where reported

Level	What it exercises	How it runs	Where reported
L1 Unit	Pure rules in `domain/`: hearts, stars, unlock, streak, grid, speech judge, prompt ladder, word bank	JUnit 4 on the JVM	§4, §5
L2 Component	ViewModels with mocked repositories, audio and recognizer (MockK, coroutine test clock); Room DAO on an in-memory database	JUnit 4, Robolectric, kotlinx-coroutines-test	§4, §5
L3 UI and layout	Compose screens on 4 screen profiles and at font scale 1.3; screen captures	Robolectric, Compose UI test, Roborazzi	§6
L4 Inspection	The built APK's permissions; the released audio against the app's files; source policies (no emoji, no haptics)	`aapt dump`, `tools/docs/check_audio_release.py`, policy tests	§5.4
L5 Device (manual)	The app on a Samsung Galaxy A21s, played as a child would	Team phone tests of APK B, D and E; acceptance check of the submitted APK, October 18	§5.5, §7.1
L6 Field	Learners, parents and teachers (Round 2)	SPMP Milestones 5–6, Weeks 7–8	§4.8 (not run)

2.2 Test Design Techniques

Technique	Where it is used
Equivalence partitioning	Say It answers split into classes: the key word, the letter's name, an added vowel, a substitution, another word, a partial word, silence (TC-ASR-07 to TC-ASR-10).
Boundary values	Hearts 5 → 0 and the 3-heart restart; recovery at the pool's cap; 6 profiles; a 16-character learner name; idle at 9 s and 10.001 s; the smallest screen, 360x640 (TC-FND-05 to 07, TC-PRF-01, TC-IND-01, §6).
State transition	The four mic states and the prompt ladder from the first to the third miss (TC-MIC-03, TC-SAY-01 to 05).
Decision table	Star rules by hearts lost (TC-PRG-01, TC-PRG-02).
Error guessing	Rapid repeated taps, the recognizer stopped by the system, a tile deal that is already solved, a stale picture after "next letter" (TC-ASR-11, TC-ASR-12, TC-BLD-02, INC-04).
Negative tests	Unknown lesson IDs, blank or too-long names, wrong and non-numeric answers at the parent gate (TC-FND-08, TC-PRF-02, TC-PZ-01).

2.3 Beyond the Happy Path

Failure mode	Test cases
Invalid inputs	TC-ASR-07 (letter names, added vowels, substitutions rejected with their error type), TC-ASR-08 (partial, misspelled and other words rejected), TC-FND-08 (unknown lesson ID), TC-PRF-02 (blank or 17-character name ignored), TC-PZ-01 (wrong and non-numeric gate answers)
Boundary conditions	TC-FND-05 (hearts reach 0), TC-FND-06 (restart with 3 hearts, losses kept), TC-FND-07 (recovery capped at the starting pool), TC-BLD-05, TC-PRF-01 (6th profile), TC-PRG-01 (star thresholds), §6 (smallest screen, font scale 1.3, 16-character name)

Failure mode	Test cases
Missing inputs and timeouts	TC-ASR-09 (blank transcript), TC-ASR-10 (no final result: judged at the 3.8 s timeout), TC-IND-01 (10 s without a touch), TC-ASR-11 (recognizer stopped by the system)
Incorrect user actions	TC-BLD-02 (tiles never dealt solved), TC-BLD-03 (wrong word: one heart, a soft sound, no buzz), TC-BLD-04 (hint after two mistakes), TC-ASR-12 (rapid taps), TC-SAY-04 (tapping the mic after the third miss does nothing)
Unauthorized access and data leaving the device	TC-PZ-01 (parent gate), TC-SEC-01 (no INTERNET permission in the APK), TC-SEC-02 (cloud backup of the database disabled)

2.4 Pass/Fail, Entry and Exit Criteria

- **Entry:** the code compiles; the run starts from clean results (`cleanTestDebugUnitTest`), so no result is reused from an earlier run.
- **A test case passes** when every cited test method passed in the recorded run, or the inspection output equals the expected result. The generator of this document refuses to build if a cited test is missing from the run, failed, or was skipped outside the allowed font-scale checks.
- **A test case fails** when the build does not do what the SRS requires, even if no automated test fails. Such a failure is reported as a defect in §7.2.
- **Exit criteria for the submission:** 0 failed automated tests; every failing test case linked to a defect with a severity and an owner; every test case that did not run listed with its reason and schedule.

2.5 Test Environment

- **Automated runs:** the team's Windows 11 development laptop; OpenJDK 17.0.16 (Eclipse Temurin), Gradle 8.9, Android Gradle Plugin 8.4.0, Kotlin 1.9.23, JUnit 4.13.2, Robolectric 4.13 emulating Android SDK 34, Roborazzi 1.26.0, MockK 1.13.10. The app is offline, so no network service is involved.
- **Screen profiles (Robolectric qualifiers):** Compact `w360dp-h640dp-xhdpi`; A21s `w360dp-h740dp-xhdpi` (the team's test phone); Phone `w411dp-h891dp-xxhdpi`; Tablet `w800dp-h1280dp-mdpi`.
- **Device:** Samsung Galaxy A21s (the team's test phone), for the manual tests in §5.5.
- **Test data:** the seeded curriculum (26 letters, 7 chapters, 33 Blend It words); fakes and mocks inside each test; no learner data.

3. Traceability: SRS → Features → Test Cases → Results

Requirement (SRS rev. 3.8)	System feature	Test cases (§4)	Result on APK H
FR-01 Carrier narration	Hear It	TC-HI-01, TC-IND-02	Pass
FR-02 Pure phoneme model and sequence	Hear It	TC-HI-01 to TC-HI-05; TC-AUD-01, TC-AUD-02, TC-SCR-01	Pass; not run: device timing, teacher audit
FR-03 Tutoring loop, mic states, prompt ladder	Say It	TC-MIC-03, TC-SAY-01 to TC-SAY-06, TC-ASR-09 to TC-ASR-12; TC-MIC-01, TC-MIC-02	Pass; not run: device timing, planned ripple
FR-NEW-REC Spaced recall	Say It	TC-REV-01 to TC-REV-03	Not run: planned (`LearnerModel`)
FR-04 Find It grid	Find It	TC-FND-01, TC-FND-02, TC-FND-09, TC-AUD-04; TC-FND-03	Pass; **Fail: TC-FND-03 (DEF-01)**

Requirement (SRS rev. 3.8)	System feature	Test cases (§4)	Result on APK H
FR-05 Hearts in Find It	Find It	TC-FND-04 to TC-FND-06, TC-FND-08	Pass
FR-07 Heart recovery	Find It	TC-FND-07	Pass
FR-13 Decodable Blend It words	Blend It	TC-BLD-01 to TC-BLD-07	Pass
FR-06 Star rules	Progression	TC-PRG-01; TC-PRG-02	Pass; **Fail: TC-PRG-02 (DEF-02)**
FR-08 Linear unlock	Progression	TC-PRG-03	Pass
FR-09 Daily streak	Progression	TC-PRG-04	Pass
FR-14 Multi-profile	Profiles	TC-PRF-01 to TC-PRF-03; TC-DB-03	Pass; partial: TC-DB-03
FR-10 Parent dashboard and gate	Parent Zone	TC-PZ-01, TC-PZ-04	Pass
FR-12 PDF report	Parent Zone	TC-PZ-02	Pass
FR-NEW-TEL Telemetry and CSV export	Parent Zone	TC-TEL-01, TC-EXP-01, TC-EXP-02	Not run: planned (Room schema v4)
NFR-AUD-01 Audio release gate	Audio	TC-AUD-03, TC-HI-03, TC-BLD-06, TC-PZ-03; TC-AUD-02	Pass; Gate 3 teacher audit pending
NFR-ASR-01 Judge accuracy and error types	Say It	TC-ASR-07, TC-ASR-08, TC-ASR-13; TC-ASR-01 to TC-ASR-06	Pass; not run: Round 2 agreement and false rejects
NFR-PERF-01 Latency	All	TC-PERF-01, TC-PERF-02	Not run: needs telemetry on the device
NFR-IND-01 Independent use, idle re-prompt	All lessons	TC-IND-01, TC-IND-02, TC-PRF-03; TC-IND-03	Pass; not run: Round 2
NFR-ACC-01 Mouth-shape cues	Say It	TC-ACC-01, TC-ACC-03, TC-SAY-03	Pass
NFR-ACC-02 64 dp targets, layout	All screens	TC-ACC-02, TC-ACC-04, TC-POL-03, layout matrix (§6)	Pass
NFR-SES-01 Session pacing	All	TC-SES-01	Not run: [Planned — Round 2], not built
NFR-REL-01 Offline persistence	Data	TC-DB-01, TC-DB-02	Not run: needs schema v4 and the device
NFR-SEC-01 Local data only	Build	TC-SEC-01, TC-SEC-02	Pass
Team policies (zero emoji, no haptics)	All screens	TC-POL-01, TC-POL-02	Pass

4. Test Case Specifications and Results

Each table gives the input, the expected result (from the SRS), the actual result observed in the run, the evidence and the status. Evidence and status come from the JUnit logs (§5).

4.1 Module 1: Letter-Sound Acquisition (Hear It)

TC ID	Req.	Type	Input / steps	Expected result	Actual result	Evidence	Status

TC-HI-01	FR-02.4, FR-01	Functional	Build the modeling sequence for M (key word "mouse")	Starts with the carrier "Listen!"; the sound 3 times with 500 ms pauses before the key word; the key word; the sound once more; ends on the sound, never "Say it with me!"	First clip `car_listen.wav`; 3 sounds before the key word, 1 after; 2 pauses of 500 ms; the last two clips are the key word and the sound; no "say it with me" clip. Loading a letter plays the whole sequence, then the next cue	`HearItSequenceBuilderTest.build_m_mouse_followsTable3`
						`HearItViewModelTest.load_playsFullModelingSequence`
						`HearItViewModelTest.firstSequenceEnd_playsNextCue_andHighlights`
							PASS (3 of 3 runs)

TC-HI-02	FR-02.6	Functional	Tap replay	Replays steps 3–6 ("This letter says…" to the sound), not the opening carrier	Replay starts at `car_this_letter_says` and ends on the sound; no "Listen!", no "say it with me"	`HearItSequenceBuilderTest.replayTemplate_isSteps3to6`
						`HearItViewModelTest.playPhonemeSound_playsReplaySegment`
						`HearItViewModelTest.replayEnd_alsoUnlocks`
							PASS (3 of 3 runs)

TC-HI-03	FR-02.1–3, NFR-AUD-01	Functional	Resolve the sound for M, S, I and the other letters	M, S and I play their released clips (`ph_m.wav`, `ph_s.wav`, `ph_i.wav`), stored apart from carriers and words; the files exist and are not empty	Paths resolve to the released WAV files, which exist with content; letters not yet remade keep the MVP clip	`AudioResolverTest.phonemeM_usesReleasedHeldSound`
						`AudioResolverTest.phonemeS_usesReleasedHeldSound`
						`AudioResolverTest.phonemeI_usesReleasedClip`
						`AudioResolverTest.releasedPhoneme_fileExists`
						`AudioResolverTest.otherPhonemes_keepOldPath`
							PASS (5 of 5 runs)

TC-HI-04	FR-02	UI	Hear It while the sound plays	One message at a time: Lily says "Listen!", the card shows the sound once, the play button keeps its colour	"Sound: /m/" appears once, the bubble says "Listen!", and the play button keeps its colour	`HearItScreenFocusTest.whilePlaying_soundTextShowsOnce_andLilySaysListen` `HearItScreenFocusTest.whilePlaying_playButtonKeepsItsColour`	**PASS** (2 of 2 runs)
TC-HI-05	FR-02	Negative	Open Hear It with an unknown or missing letter ID	No crash: an unknown ID sets a load error; a missing ID falls back to letter 1	Load error set for an unknown ID; a missing ID loads letter 1	`HearItViewModelTest.loadPhoneme_invalidId_setsLoadError` `HearItViewModelTest.loadPhoneme_nullId_defaultsToId1`	**PASS** (2 of 2 runs)

4.2 Module 2: Speech Production and Tutoring (Say It)

TC ID	Req.	Type	Input / steps	Expected result	Actual result	Evidence	Status
TC-MIC-03	FR-03.2	State transition	Map each Say It state to the mic	Four distinct states, none red: Idle, Listening, Heard (speech detected), Result (correct or try again); after the third miss the mic rests	Idle → IDLE; listening without speech → LISTENING; with speech → HEARD; correct → RESULT_CORRECT; wrong → RESULT_TRY_AGAIN; after the third miss → DONE	`MicStatusTest` (all 6)	**PASS** (6 of 6 runs)
TC-SAY-01	FR-03.4	State transition	First miss: the child says the letter's name ("em")	Spoken correction for the error type: "That's the letter's name. Its sound is /m/", re-model, "Your turn"	Plays the soft sound, `fb_letter_name`, `fb_its_sound_is`, the sound, the word and `car_your_turn`; tutor action is a level-1 correction of type LETTER_NAME; Next stays locked	`SayItViewModelTest.firstLetterNameMiss_playsLetterNameCorrection` `TutorPolicyTest.firstMiss_correctsAtLevel1_withErrorType`	**PASS** (2 of 2 runs)
TC-SAY-02	FR-03.4	State transition	First miss: an added vowel ("ma"); or another word	Added vowel: "Almost! Just /m/, no 'ah'" with a re-model; another word: "Listen" and a re-model	The added-vowel correction (`fb_no_ah`) plays; another word gets the "Listen" re-model	`SayItViewModelTest.addedVowelError_triggersFbNoAhSpokenCorrection` `SayItViewModelTest.otherWordMiss_playsListenRemodel`	**PASS** (2 of 2 runs)
TC-SAY-03	FR-03.4, NFR-ACC-01	State transition	Second miss	Slower re-model with the mouth-shape cue	No cue after the first miss; the cue shows after the second, with "Watch my lips"; level-2 correction	`SayItViewModelTest.secondMiss_showsMouthCue` `SayItViewModelTest.secondMiss_playsWatchMyLips` `TutorPolicyTest.secondMiss_correctsAtLevel2`	**PASS** (3 of 3 runs)

TC ID	Req.	Type	Input / steps	Expected result	Actual result	Evidence	Status
TC-SAY-04	FR-03.4	State transition, negative	Third miss, then tap the mic again	Together-practice ("Let's say it together"), the lesson moves on without a failure screen, the miss is saved; the mic does not reopen	Plays `car_lets_say_together`, the word, `fb_try_later`; Next unlocks; 3 incorrect attempts saved; the mic shows DONE; a later tap does not start the recognizer	`SayItViewModelTest.thirdMiss_emitsLeadAndMoveOn_andSavesIncorrect` `SayItViewModelTest.thirdMiss_micRestsAndNextUnlocks` `SayItViewModelTest.afterLeadAndMoveOn_startListeningIsIgnored` `TutorPolicyTest.thirdMiss_leadsAndMovesOn`	**PASS** (4 of 4 runs)
TC-SAY-05	FR-03.5	Boundary	Wrong answers, including three in a row	Say It never removes a heart	Hearts after a wrong answer and after the third miss equal the hearts before	`SayItViewModelTest.evaluateSpeech_incorrectTranscript_doesNotDeductHeart` `SayItViewModelTest.thirdMiss_emitsLeadAndMoveOn_andSavesIncorrect`	**PASS** (2 of 2 runs)
TC-SAY-06	FR-03	Functional	Say the key word, first try or after a miss	Praise ("Yes!") and Next unlocks	State Correct; praise action; Next available after a miss too	`SayItViewModelTest.evaluateSpeech_correctTranscript_setsCorrectState` `SayItViewModelTest.correctAfterMiss_praises_andCanContinue` `TutorPolicyTest.correct_anyAttempt_praises`	**PASS** (3 of 3 runs)
TC-ASR-07	NFR-ASR-01.4	Invalid input	Foils: "em", "es", "bee", "zee", "ay", "m"; "ma", "muh", "sa", "buh"; "pa" for /f/, "ba" for /v/, "sa" for /z/	Every foil is rejected and labelled LETTER_NAME, ADDED_VOWEL or SUBSTITUTION	All 13 foils rejected with the expected error type; the ViewModel shows the matching state	`SpeechValidatorTest.judgeSound_letterName_rejected` `SpeechValidatorTest.judgeSound_addedVowel_rejected` `SpeechValidatorTest.judgeSound_substitution_rejected` `SayItViewModelTest.evaluateSpeech_letterName_setsIncorrectWithLetterName` `SayItViewModelTest.evaluateSpeech_addedVowel_setsIncorrectWithAddedVowel`	**PASS** (5 of 5 runs)
TC-ASR-08	FR-03.6	Invalid input, equivalence	Word mode: the 26 key words; and "m", "muh", "mmm", "mous", "elphant", "sunny", "sub" for "sun", "box" for "fox"	Each key word is accepted (any case or spacing); letter sounds, partial or misspelled words and other words are rejected	26 of 26 key words accepted; all wrong forms rejected; the grammar holds the word and its foils, not generic decoys	`SpeechValidatorTest.wordExactMatch_returnsTrue_forAll26ExampleWords` `SpeechValidatorTest.wordMode_caseAndWhitespaceInsensitive` `SpeechValidatorTest.wordMode_rejectsLetterSoundOnly` `SpeechValidatorTest.wordMode_rejectsPartialOrMisspelledWords` `SpeechValidatorTest.wordMode_`	**PASS** (6 of 6 runs)

TC ID	Req.	Type	Input / steps	Expected result	Actual result	Evidence	Status
						`rejectsOtherWordsAndMinimalPairs` `SpeechValidatorTest.grammarFor_ wordMode_hasFoilsNotGenericDecoys`	
TC-ASR-09	FR-03.4	Missing input	Null, empty or blank transcript	Not accepted (counts as a miss)	Rejected in both letter and word mode	`SpeechValidatorTest.nullOrBlankText_ returnsFalse` `SpeechValidatorTest.wordMode_ nullOrBlankTranscript_returnsFalse`	**PASS** (2 of 2 runs)
TC-ASR-10	FR-03.4	Timeout	A wrong partial result ("em", "ma") with no final result	A partial does not end the attempt (it may be the start of the word); without a final result the attempt is judged at the timeout	After wrong partials the mic keeps listening and nothing is scored; at the 3.8 s auto-stop "ma" is judged ADDED_VOWEL; a partial followed by the right word is correct	`SayItViewModelTest.onResult_ partialFoil_keepsListening` `SayItViewModelTest.onResult_ partialAddedVowelThenTimeout_ setsIncorrectWithAddedVowel` `SayItViewModelTest.onResult_ partialAddedVowelThenFinalWord_ isCorrect` `SayItViewModelTest.onResult_ partialLetterNameThenFinalWord_ isCorrect`	**PASS** (4 of 4 runs)
TC-ASR-11	FR-03	Error guessing	The system stops the recognizer while listening (app sent to the background)	Return to Idle; the dropped attempt is not scored	Mic IDLE, state Idle, no attempt recorded, and the cancelled timeout does not judge it later	`SayItViewModelTest. recognizerStoppedExternally_ returnsToIdle`	**PASS** (1 of 1 runs)
TC-ASR-12	FR-03	Incorrect action	Tap "hear the word" three times quickly	The word plays once	One playback	`SayItViewModelTest.playWordAudio_ rapidTaps_debounced`	**PASS** (1 of 1 runs)
TC-ASR-13	FR-03.6	Configuration	Default scoring mode	Pure-sound scoring is off (the Vosk spike could not confirm a pure sound); a held sound alone is never accepted while off	Sound mode off by default; a sustained sound is "unconfirmed"; the sound-mode grammar accepts nothing	`SpeechValidatorTest.soundMode_ isOffByDefault` `SpeechValidatorTest.judgeSound_ soundModeOff_sustainedIsUnconfirmed` `SpeechValidatorTest.grammarFor_ soundMode_hasNoAcceptedTokens`	**PASS** (3 of 3 runs)

4.3 Module 3: Phoneme Discrimination (Find It)

TC ID	Req.	Type	Input / steps	Expected result	Actual result	Evidence	Status

TC ID	Req.	Type	Input / steps	Expected result	Actual result	Evidence	Status
TC-FND-01	FR-04	Functional	Open Find It for M	5 cards: exactly 3 targets and 2 distractors	5 cards, 3 correct and 2 distractors	`GridGeneratorTest.generate5ItemGrid_returnsExactly5ItemsWith3CorrectAnd2Distractors` `FindItViewModelTest.loadGrid_validId_loadsTargetAnd5ItemGrid`	**PASS** (2 of 2 runs)
TC-FND-02	FR-04	Functional	Generate 200 grids for every letter	No distractor starts with the target's sound (c, k and q count as one sound); no distractor from the x, ng or ñ banks	0 violations	`GridGeneratorTest.distractors_neverShareTheTargetSound` `GridGeneratorTest.distractorLettersFor_k_excludesSameSoundGroup` `GridGeneratorTest.distractors_neverFromSpecialBanks`	**PASS** (3 of 3 runs)
TC-FND-03	FR-04	Functional	Open Find It for letter 1 (M) and letter 2 (S) on a new profile	Distractors come "strictly from previously mastered letters" (SRS FR-04; `01_REQUIREMENTS_SUMMARY.md`)	**Not met.** `generate5ItemGrid` ignores the list of letters it is given and draws distractors from every curriculum letter. The captures in Figure 3 show M with Elephant, Orange, Rabbit, Sun, Jug, Yoyo and Pig, none of them taught yet. No test covers this rule.	Code inspection: `GridGenerator.kt` lines 62–98; `FindItViewModel.kt` line 123; Figure 3	**FAIL (DEF-01)**
TC-FND-04	FR-05	Functional	Tap a distractor	Exactly 1 heart lost, a gentle sound, the miss saved	Hearts 5 → 4; state Incorrect; attempt saved as incorrect	`FindItViewModelTest.selectPictureItem_incorrectItem_deductsHeart` `HeartManagerTest.deductHeart_decrementsCountAndIncrementsHeartsLost`	**PASS** (2 of 2 runs)
TC-FND-05	FR-05	Boundary	5 wrong taps; then one more deduction	At 0 hearts the activity stops (game over); hearts never go below 0	Hearts 0, state GameOver; a further deduction keeps 0	`FindItViewModelTest.selectPictureItem_allHeartsLost_setsGameOver` `HeartManagerTest.deductHeart_signalsGameOverWhenZero`	**PASS** (2 of 2 runs)

TC ID	Req.	Type	Input / steps	Expected result	Actual result	Evidence	Status
TC-FND-06	FR-05	Boundary	Restart after depletion	Reload with a 3-heart buffer; the hearts lost before the restart still count for stars	Hearts 3, found count 0, not game over; the session's hearts lost survive the restart	`FindItViewModelTest.restartSession_resetsHeartsAndState` `HeartManagerTest.restartAtReducedPoolOnDepletion` `FindItViewModelTest.sessionHeartsLost_survivesRestart` `HeartManagerTest.sessionHeartsLost_survivesRestart` `FindItViewModelTest.wrongTaps_countIntoSessionHeartsLost`	**PASS** (5 of 5 runs)
TC-FND-07	FR-07	Boundary	1 wrong tap, then 3 right taps in a row; and 3 right taps at a full pool of 3	+1 heart per 3 in a row, never above the session's starting pool	Hearts 4 → 5 after three in a row; a full pool of 3 stays 3	`FindItViewModelTest.threeCorrectInARow_recoversAHeart` `HeartManagerTest.checkRecovery_addsHeartEveryThreeConsecutiveCorrect` `HeartManagerTest.recoveryCapsAtStartingPool_recommendedRule`	**PASS** (3 of 3 runs)
TC-FND-08	FR-05	Negative	Open Find It, Say It or Letter Complete with an unknown letter ID (999)	No crash; a load error is shown	Target null and load error set; same in Say It and Letter Complete	`FindItViewModelTest.loadGrid_invalidId_setsLoadError` `SayItViewModelTest.loadPhoneme_invalidId_setsLoadError` `LetterCompleteViewModelTest.completeLesson_handlesMissingPhoneme_andSetsLoadError`	**PASS** (3 of 3 runs)
TC-FND-09	FR-04	Functional	Find all 3 targets	The lesson completes and the next cue plays	Found count 3, lesson complete, next cue appended	`FindItViewModelTest.selectPictureItem_correctItem_incrementsFoundCount` `FindItViewModelTest.selectPictureItem_allThreeCorrect_completesLesson` `FindItViewModelTest.completion_appendsNextCue`	**PASS** (3 of 3 runs)

4.4 Module 4: Word Construction (Blend It)

TC ID	Req.	Type	Input / steps	Expected result	Actual result	Evidence	Status

TC ID	Req.	Type	Input / steps	Expected result	Actual result	Evidence	Status
TC-BLD-01	FR-13.1–2	Data validation	Check every seeded word against the letters taught by its group	100% decodable with the letters taught so far; AIM, BEE, TOY, BOY, ZOO gone; AM, SUM, TUB, YAM, ZIP present in the same slots; QUIZ the one documented exception	33 words (3 in group 1, then 5 per group); 0 untaught letters except QUIZ; the 5 old words absent and the 5 new ones present	`BlendItWordSeedsTest.everyWordUsesOnlyTaughtLetters` `BlendItWordSeedsTest.nonDecodableWordsAreGone` `BlendItWordSeedsTest.replacementsKeepTheirSlots` `BlendItWordSeedsTest.thirtyThreeWords_threeThenFivePerGroup` `BlendItWordSelectorTest` (all 2)	**PASS** (6 of 6 runs)
TC-BLD-02	FR-13.3	Incorrect action	Deal the tiles 200 times for SAM, SIS, AM, TUB, YAM, ZIP, QUIZ, MOM	Never dealt in the answer's order; always the same letters	1,600 deals, none solved, none with a letter lost or added; a word with one possible order (MM) keeps it	`BlendItTileDealTest` (all 2)	**PASS** (2 of 2 runs)
TC-BLD-03	FR-13, FR-05	Incorrect action	Submit M-A-S for SAM	One heart lost; a soft "not yet" sound, never a buzz; the miss saved	Hearts −1; 1 wrong attempt; state WordIncorrect; the pop plays and the buzz does not	`BlendItViewModelTest.submitWord_incorrectWord_deductsHeart` `BlendItViewModelTest.wrongSubmit_playsSoftPop_notBuzz`	**PASS** (2 of 2 runs)
TC-BLD-04	FR-13	Incorrect action	Two wrong submissions, then the hint	The first letter is locked in place	1 locked hint; the first placed tile is S	`BlendItViewModelTest.applyHint_afterTwoMistakes_locksCorrectCharacter`	**PASS** (1 of 1 runs)
TC-BLD-05	FR-13, FR-05	Boundary	5 wrong submissions, then restart	At 0 hearts the depleted state shows; the restart gives 3 hearts and keeps the 5 lost for stars	Hearts 0 → HeartDepleted → restart: 3 hearts, word 1, Idle; result reports 5 hearts lost	`BlendItViewModelTest.heartDepleted_restartsWithThreeHearts`	**PASS** (1 of 1 runs)
TC-BLD-06	FR-13, NFR-AUD-01	Functional	Resolve the AM, TUB and YAM word clips	The released WAV clips (audio release `2026-10-11-words`) exist and are used	The three WAV files exist with content; the seeds point at them	`AudioResolverTest.releasedWords_filesExist` `AudioResolverTest.releasedWords_useWav` `BlendItWordSeedsTest.releasedWordsPointAtTheirWav`	**PASS** (3 of 3 runs)

TC ID	Req.	Type	Input / steps	Expected result	Actual result	Evidence	Status
TC-BLD-07	FR-13	Functional	A correct word	Each tile's sound plays in turn before the whole word; the next word follows	Every tile lights and sounds before the word; the word advances	`BlendItViewModelTest.soundOut_waitsForEachSound` `BlendItViewModelTest.soundOut_lightsAllTilesBeforeWord` `BlendItViewModelTest.submitWord_correctWord_transitionsToWordCorrectAndAdvances`	**PASS** (3 of 3 runs)

4.5 Module 5: Progression, Gamification and Profiles

TC ID	Req.	Type	Input / steps	Expected result	Actual result	Evidence	Status
TC-PRG-01	FR-06	Decision table	Finish a letter with 0, 1, 2, 3 and 4 hearts lost; carry the count through navigation	0 lost: 3 stars; 1–2 lost: 2 stars; 3 or more: 1 star; the count reaches Letter Complete; 1 star says "You finished", not "mastered"	0 → 3, 1 → 2, 2 → 2, 3 → 1, 4 → 1; the route carries the count; the 1-star message is "You finished Letter M!"	`StarCalculatorTest` (all 3) `LetterCompleteViewModelTest.completeLesson_calculates3StarsFor0HeartsLost_andSavesProgress` `LetterCompleteViewModelTest.completeLesson_calculates2StarsFor1HeartLost` `LetterCompleteViewModelTest.heartsLostArg_threeOrMore_givesOneStar` `RoutesTest.letterComplete_carriesHeartsLost` `LetterCompleteMessageTest` (all 2)	**PASS** (9 of 9 runs)
TC-PRG-02	FR-06	Decision table	Find It with 1 wrong tap (3 correct of 4 taps = 75% accuracy)	SRS FR-06: 2 stars need at least 80% accuracy, so 75% gives 1 star	**Not met.** Letter Complete passes only hearts lost to `StarCalculator`, whose accuracy defaults to 100%, so this case gets 2 stars; a test (`completeLesson_calculates2StarsFor1HeartLost`) expects 2. With 3 targets, 1 wrong tap is always 75%, so the SRS rule as written never gives 2 stars in Find It.	Code inspection: `LetterCompleteViewModel.kt` line 102, `StarCalculator.kt`	**FAIL (DEF-02)**

TC ID	Req.	Type	Input / steps	Expected result	Actual result	Evidence	Status
TC-PRG-03	FR-08	Functional	Letter 2 with letter 1 incomplete, then complete	Letter N+1 stays locked until letter N is complete; Blend It opens when its group is complete	Locked, then unlocked; group and Blend It unlock follow the same rule	`UnlockManagerTest` (all 3) `GroupUnlockManagerTest` (all 4) `MapTapRulesTest.lockedNode_staysLocked`	**PASS** (8 of 8 runs)
TC-PRG-04	FR-09	Boundary	Activity on the first day, the same day, the next day, after 50 hours	First activity: streak 1; same day: unchanged; next day: +1; after a missed day: back to 1 on activity, 0 at the inactivity reset, stars untouched; badges at 5, 10, 15, 20 days	All as expected; 15 stars kept after the reset	`StreakTrackerTest` (all 7)	**PASS** (7 of 7 runs)
TC-PRF-01	FR-14.1	Boundary	5 profiles, then 6	A new profile can be added below 6 and not at 6	`canAddProfile` true at 5, false at 6	`ProfileViewModelTest.canAddProfile_returnsTrueWhenUnderLimit_andFalseWhenAtMax`	**PASS** (1 of 1 runs)
TC-PRF-02	FR-14	Invalid input	Rename to " " and to a 17-character name; rename with spaces around	Blank or longer than 16 characters is ignored; a valid name is trimmed	No update for the two invalid names; the valid one saved trimmed	`ParentDashboardViewModelTest.renameProfile_blankOrTooLong_isIgnored` `ParentDashboardViewModelTest.renameProfile_updatesTrimmedName`	**PASS** (2 of 2 runs)

TC ID	Req.	Type	Input / steps	Expected result	Actual result	Evidence	Status
TC-PRF-03	FR-14, NFR-IND-01	Functional	Create a profile without typing a name; delete the active, another and the last profile	The child never has to type (the avatar's name is used); deleting the active profile ends its session; deleting the last shows "no profiles"	Avatar name used for a blank name; sessions and the "no profiles" state as expected	`ProfileViewModelTest.createProfile_blankName_usesAvatarName` `ParentDashboardViewModelTest.deleteActiveProfile_clearsSession` `ParentDashboardViewModelTest.deleteOtherProfile_keepsSession` `ParentDashboardViewModelTest.deleteLastProfile_setsNoProfilesLeft`	**PASS** (4 of 4 runs)
TC-DB-03	FR-14.2, 14.4	Data isolation	Two profiles with their own lesson and Blend It progress (in-memory Room database)	Each profile's totals count only its own progress	Ana 2 stars, Ben 4 stars, in both queries. Only star totals are covered; concurrent writes and the other per-profile tables are not	`ProfileDaoStarsTest` (all 4)	**PARTIAL** (the 4 runs pass; scope incomplete)

4.6 Module 6: Parent Zone

TC ID	Req.	Type	Input / steps	Expected result	Actual result	Evidence	Status
TC-PZ-01	FR-10	Unauthorized access	Answer the arithmetic gate with the right answer (with spaces), a wrong one, blank, "abc", "12a"	Only the right number opens the Parent Zone	Right answer (trimmed) accepted; the other four rejected	`ArithmeticGateManagerTest` (all 4)	**PASS** (4 of 4 runs)
TC-PZ-02	FR-12	Functional	Export the PDF report; and an export that fails	Success and failure are both reported to the parent; no crash	Export status set for both	`ParentDashboardViewModelTest.exportPdfReport_success_updatesExportStatus` `ParentDashboardViewModelTest.exportPdfReport_failure_updatesExportStatus`	**PASS** (2 of 2 runs)

TC ID	Req.	Type	Input / steps	Expected result	Actual result	Evidence	Status
TC-PZ-03	NFR-AUD-01.1	Compliance	Read the Parent Zone credit	Names ElevenLabs and "elevenlabs.io" (Free-plan terms for the AM, TUB, YAM and /i/ clips)	Credit present with the domain	`VoiceCreditTest.creditNamesElevenLabsAndItsDomain`	**PASS** (1 of 1 runs)
TC-PZ-04	FR-10	Functional	Open the dashboard with profiles, with none, and switch profile	The first profile's data loads; an empty list ends loading; switching loads the selected data; errors are handled	All four as expected	`ParentDashboardViewModelTest.init_selectsFirstProfileAndLoadsDashboardData` `ParentDashboardViewModelTest.init_withEmptyProfiles_setsLoadingFalse` `ParentDashboardViewModelTest.selectProfile_loadsDataForSelectedProfile` `ParentDashboardViewModelTest.selectProfile_handlesDashboardDataError`	**PASS** (4 of 4 runs)

4.7 Cross-Cutting Requirements

TC ID	Req.	Type	Input / steps	Expected result	Actual result	Evidence	Status
TC-IND-01	NFR-IND-01.4	Timeout	No touch for 9 s, touch, then 10 s; audio still playing; screen hidden	Re-prompt after 10 s without input; a touch restarts the clock; no re-prompt while Lily is speaking or the screen is hidden; at most 3 re-prompts	No nudge at 9 s after a touch; 1 nudge at 10.001 s; none during 30 s of playing audio; none when hidden; stops after the maximum; same in Hear It, Find It and Blend It	`IdleTimerTest` (all 6) `SayItViewModelTest.idle_tenSecondsWithoutTouch_nudgesOnce` `SayItViewModelTest.idle_waitsWhileTheModelPlays` `SayItViewModelTest.idle_noNudgeOnceHidden` `HearItViewModelTest.idle_nudgesTheScreen` `FindItViewModelTest.idle_replaysIntro` `BlendItViewModelTest.idle_replaysIntro`	**PASS** (12 of 12 runs)
TC-IND-02	NFR-IND-01.2, FR-01	Completeness	Look for every spoken prompt the lessons use	Every carrier, key word, Say It word prompt and lesson line exists on disk, so no step is silent	All required files present: the 26 key words, the Say It prompts, the approved tutor carriers, the lesson lines in WAV	`AudioCompletenessCheckTest` (all 6) `AudioResolverTest.lessonVo_allReleasedLinesUseWav`	**PASS** (7 of 7 runs)

TC ID	Req.	Type	Input / steps	Expected result	Actual result	Evidence	Status
TC-ACC-01	NFR-ACC-01.1	Completeness	Look up the mouth shape for all 26 letters	Every letter has a mouth-shape cue, and the picture is released	26 of 26 mapped to the right group (lips together, teeth on lip, no round lips for short vowels); every cue's picture is in an image release	`ArticulationGroupTest` (all 8) `PictureAssetsTest.everyMouthCueIsReleased` `ArticulationCueTest` (all 2)	**PASS** (11 of 11 runs)
TC-ACC-03	NFR-ACC-01	Error guessing	Move to the next letter on a lesson screen	The new letter's picture shows, not the last one; a missing picture is blank, not a crash	The new picture replaces the old; a missing asset renders transparent	`AssetImageLoadTest` (all 3)	**PASS** (3 of 3 runs)
TC-ACC-02	NFR-ACC-02	Boundary	Measure tiles, mic, main buttons, map buttons and dialog closes; map text sizes	Every child-facing target at least 64 x 64 dp; text at least 16 sp	Design tokens at least 64 dp on all profiles; map back and guidebook buttons and the two dialog closes at least 64 dp; no map text under 16 sp; the header card is Lily's 64 dp tap target	`DimensTest` (all 5) `LayoutMatrixTest.map_buttonsAre64dp` (4 sizes) `TopStatsBarLayoutTest.nodeDialog_closeIs64dp` `TopStatsBarLayoutTest.guidebookDialog_closeIs64dp` `TopStatsBarLayoutTest.card_isLilysTapTarget` `MapTextFloorTest.noMapTextUnder16sp`	**PASS** (13 of 13 runs)
TC-ACC-04	NFR-ACC-02	Boundary	A 16-character learner name on the map header	The greeting, letter count and pills are not cut	Not ellipsized; pills visible	`TopStatsBarLayoutTest.longName_greetingIsNotCut` `TopStatsBarLayoutTest.longName_pillsStayVisible` `TopStatsBarLayoutTest.lettersCount_isNotCut` `MapTapRulesTest.greeting_fitsOneLine`	**PASS** (4 of 4 runs)
TC-POL-03	NFR-ACC-02	Accessibility	Reduced motion on (system setting)	Animations become fades or stop; nothing moves by itself	Splash, motion tokens and heart wobble are still with reduced motion	`SplashScreenTest.motionPolicy_followsReducedMotion` `PlayItMotionTest.reducedMotionSnaps` `FeedbackEffectsTest.heartWobble_reducedMotionIsStill`	**PASS** (3 of 3 runs)

TC ID	Req.	Type	Input / steps	Expected result	Actual result	Evidence	Status
TC-AUD-03	NFR-AUD-01.4	Inspection	Compare every clip in the 7 audio release manifests with the app's file (SHA-256)	Every released clip the app ships is byte-identical to its release	76 released clips: 69 identical (PASS), 0 different, 7 approved but not used by APK H (no reference in the code). 159 shipped files predate the release gate (the MVP clips the SRS lists as pending)	`docs/std/evidence/2026-10-14-0320385/audio_release_check.txt` (`tools/docs/check_audio_release.py`)	**PASS** (69 of 69 shipped clips)
TC-AUD-04	FR-04, NFR-ACC-01	Inspection	Compare every released picture with the app's file (SHA-256); every grid picture exists	Identical bytes; no missing picture	All identical; all grid pictures present	`PictureAssetsTest.releasedPicturesMatchManifest` `PictureAssetsTest.everyGridPictureExists`	**PASS** (2 of 2 runs)
TC-SEC-01	NFR-SEC-01	Inspection	`aapt dump permissions` on the submitted APK	No `android.permission.INTERNET`; only the microphone (and legacy storage, capped at old Android versions)	Permissions: `RECORD_AUDIO`; `READ_EXTERNAL_STORAGE` (max SDK 32); `WRITE_EXTERNAL_STORAGE` (max SDK 28); the app's own receiver permission. **No INTERNET**	`docs/std/evidence/2026-10-18-submitted-apk/apk_inspection.txt` (October 18, the submitted file); the same permissions on October 14 (`docs/std/evidence/2026-10-14-0320385/apk_inspection.txt`, the earlier build of the same commit)	**PASS**
TC-SEC-02	NFR-SEC-01	Inspection	Read the manifest and backup rules	Learner data is not copied off the device by Android backup	`allowBackup="false"`; the backup rules exclude the database (and all other app data); the data-extraction rules name the database	`ManifestPrivacyTest.backupIsDisabled`	**PASS** (1 of 1 runs)
TC-POL-01	Team policy	Inspection	Scan every `.kt` and `.xml` file in `app/src/main`	No emoji anywhere in the app	0 violations	`ZeroEmojiPolicyTest.noEmojiInAppSources`	**PASS** (1 of 1 runs)
TC-POL-02	Team policy	Inspection	Scan the code	No haptics; portrait only; no endless idle animation on the letter card or mascot header	0 haptic calls; portrait locked; no endless idle motion	`PerformancePolicyTest` (all 4)	**PASS** (4 of 4 runs)

4.8 Test Cases Not Run on This Build

TC ID	Req.	What it checks	Why it did not run	When and how (SPMP)
TC-AUD-01	FR-02	SoundPool start latency ≤ 50 ms	Needs instrumented timing on the phone	Device test before Round 2 (Milestone 5)
TC-AUD-02	FR-02, NFR-AUD-01.3	Zero schwa: 26 of 26 sounds rated "Pure" by ≥ 3 of 4 teachers	Gate 3 teacher audit not yet held; Chapter 1's short /a/ and Chapters 2–7 still use MVP clips	SPMP Milestone 4 (Week 6) planned the audit; not yet held. Instrument A, Part 1
TC-SCR-01	FR-02.5	Modeling sequence ≤ 15 s ([Planned — Round 2])	Needs timing on the device	Instrument D (D-11), Round 2
TC-MIC-01	FR-03.3, NFR-PERF-01.3	Tap-to-Listening ≤ 100 ms	Needs timing on the device	Device test (Instrument D)
TC-MIC-02	FR-03.2	Ripple follows the voice's loudness	Planned (`AudioRecord` loop, SDD §3.2); the ripple is time-based in APK H	After Round 2 (team decision 2026-10-06)
TC-ASR-01 to 05	NFR-ASR-01.1, 3	≥ 80% agreement with teachers across child voices and classroom noise	Needs recorded child attempts and teacher ratings	Instruments B and C, Round 2 (Weeks 7–8)
TC-ASR-06	NFR-ASR-01.2	False rejects ≤ 15% of teacher-correct attempts	As above	Round 2
TC-PERF-01, 02	NFR-PERF-01	Say It P90 ≤ 0.5 s; Find It P90 ≤ 0.3 s	Needs the telemetry log (planned) on the device	After Room schema v4; Instrument D (D-01, D-02)
TC-IND-03	NFR-IND-01.1	≥ 85% of lessons completed without adult help	Field measure	Silent-Observer protocol, Round 2
TC-REV-01 to 03	FR-NEW-REC	Warm-up and end-of-session recall checks	`LearnerModel` and the review scheduler are planned (SDD §2.0)	After Round 2
TC-TEL-01	FR-NEW-TEL	Telemetry schema	`TelemetryEventEntity` and `TelemetryLogger` are planned (Room schema v4)	SPMP Milestone 4 (Week 6) planned it; not in APK H, to be re-scheduled
TC-EXP-01, 02	FR-NEW-TEL.4	PIN-gated CSV export; zero network transmission	`CsvExportManager` is planned. (The app already has no INTERNET permission: TC-SEC-01.)	SPMP Milestone 4 (Week 6) planned it; not in APK H, to be re-scheduled
TC-DB-01	NFR-REL-01	20 force-close tests with no progress lost	Needs the device and the planned transactional DAOs	Instrument D (D-05, D-06)
TC-DB-02	NFR-REL-01	Migration from schema v3 to v4 keeps data	Schema v4 is planned; `exportSchema` and explicit migrations are prerequisites (SDD §6)	With schema v4
TC-SES-01	NFR-SES-01	Session ends at the first completed node after about 12 minutes	[Planned — Round 2]; not built in APK H	Round 2 implementation

5. Test Execution

5.1 Execution Record

Item	Record
Test item	PlayIT debug APK H: branch `refactor/hear-say-it`, app code at commit `0320385` (the run used checkout `e2b6d4b`; no file under `app/` changed after `0320385`)
Command	`gradlew cleanTestDebugUnitTest testDebugUnitTest` (all results regenerated, nothing reused)
Executed	October 14, 2026, 22:13:37 to 22:14:42 (UTC+8); 63.0 s inside test classes
Executed by	Capstone Team on the team's development workstation
Environment	OpenJDK 17.0.16 (Eclipse Temurin); Gradle 8.9; Android Gradle Plugin 8.4.0; Kotlin 1.9.23; JUnit 4.13.2; Robolectric 4.13 (Android SDK 34); Roborazzi 1.26.0; MockK 1.13.10; kotlinx-coroutines-test 1.8.0
Test classes	66
Test cases run	433
Passed	427
Failed or errors	0
Skipped	6 (font-scale 1.3 checks on the 360x640 profile, by design: that profile at 1.3 is below the supported floor)
Raw logs	`docs/std/evidence/2026-10-14-0320385/junit/` (66 JUnit XML files, as written by Gradle)

5.2 Results by Test Class

Every test class of the suite, the feature and requirement it verifies, and its result. The per-test log is in Appendix A.

Test class	Feature	Requirement	Tests	Result	Time (s)
`HearItSequenceBuilderTest`	Hear It: modeling sequence	FR-02	4	4 passed	0.01
`HearItViewModelTest`	Hear It: playback, replay, next cue	FR-01, FR-02, NFR-IND-01	14	14 passed	1.37
`HearItScreenFocusTest`	Hear It: one message at a time	FR-02	2	2 passed	2.99
`HearItScreenshotTest`	Hear It: screen capture	NFR-ACC-02	1	1 passed	0.27
`AudioResolverTest`	Audio: released clips and paths	FR-02, FR-13, NFR-AUD-01	23	23 passed	0.06
`AudioCompletenessCheckTest`	Audio: every required clip on disk	NFR-AUD-01, NFR-IND-01	6	6 passed	0.04
`CaptionTextTest`	Audio: caption text (kept in code)	NFR-ACC-01	4	4 passed	0.00
`SayItViewModelTest`	Say It: tutoring loop, judge, idle	FR-03, NFR-ASR-01, NFR-IND-01	38	38 passed	2.87
`SayItFeedbackCopyTest`	Say It: feedback wording	FR-03	8	8 passed	0.04

Test class	Feature	Requirement	Tests	Result	Time (s)
MicStatusTest	Say It: four mic states	FR-03	6	6 passed	0.00
TutorPolicyTest	Say It: prompt ladder	FR-03	4	4 passed	0.00
SpeechValidatorTest	Say It: word judge and foils	NFR-ASR-01, FR-03	27	27 passed	0.03
ArticulationGroupTest	Say It: letter-to-mouth pairs	NFR-ACC-01	8	8 passed	0.01
ArticulationCueTest	Say It: mouth-shape picture	NFR-ACC-01	2	2 passed	5.49
FindItViewModelTest	Find It: taps, hearts, restart	FR-04, FR-05, FR-07	12	12 passed	0.80
GridGeneratorTest	Find It: grid and distractors	FR-04	7	7 passed	0.36
HeartManagerTest	Hearts: deduct, restart, recover	FR-05, FR-07	9	9 passed	0.00
FindItScreenshotTest	Find It: screen capture	NFR-ACC-02	1	1 passed	0.38
PictureAssetsTest	Pictures: released images	FR-04, NFR-ACC-01	3	3 passed	0.45
BlendItViewModelTest	Blend It: tiles, hints, hearts	FR-13, FR-05	11	11 passed	1.38
BlendItTileDealTest	Blend It: tile deal	FR-13	2	2 passed	0.03
BlendItWordSeedsTest	Blend It: decodable word bank	FR-13	5	5 passed	0.01
BlendItWordSelectorTest	Blend It: words per group	FR-13	2	2 passed	0.00
BlendItCompleteViewModelTest	Blend It: group stars	FR-06, FR-13	4	4 passed	4.88
BlendItStarThresholdsTest	Blend It: star thresholds	FR-06	3	3 passed	0.00
BlendItCardLayoutTest	Blend It: card layout	NFR-ACC-02	1	1 passed	0.95
BlendItScreenshotTest	Blend It: screen capture	NFR-ACC-02	1	1 passed	0.57
StarCalculatorTest	Progress: letter stars	FR-06	3	3 passed	0.01
LetterCompleteViewModelTest	Progress: letter complete	FR-06	9	9 passed	0.89
LetterCompleteMessageTest	Progress: completion message	FR-06	2	2 passed	0.00
LetterCompleteScreenshotTest	Progress: screen capture	NFR-ACC-02	1	1 passed	0.19
RoutesTest	Progress: results carried by navigation	FR-06	4	4 passed	0.01
UnlockManagerTest	Progress: linear unlock	FR-08	3	3 passed	0.00
GroupUnlockManagerTest	Progress: group and Blend It unlock	FR-08	4	4 passed	0.00
StreakTrackerTest	Progress: daily streak	FR-09	7	7 passed	0.13
LetterStatusCalculatorTest	Parent Zone: letter status flags	FR-10	4	4 passed	0.00
RetentionCalculatorTest	Parent Zone: retention score	FR-10	2	2 passed	0.00

Test class	Feature	Requirement	Tests	Result	Time (s)
ParentDashboardViewModelTest	Parent Zone: dashboard, report, profiles	FR-10, FR-12, FR-14	13	13 passed	0.84
ArithmeticGateManagerTest	Parent Zone: arithmetic gate	FR-10	4	4 passed	0.00
VoiceCreditTest	Parent Zone: ElevenLabs credit	NFR-AUD-01	1	1 passed	0.00
ProfileViewModelTest	Profiles: create, select, limit	FR-14, NFR-IND-01	10	10 passed	0.53
ProfileDaoStarsTest	Profiles: per-profile star totals	FR-14, FR-06	4	4 passed	6.80
NamePromptScreenshotTest	Profiles: screen capture	NFR-ACC-02	1	1 passed	0.13
MapViewModelTest	Map: nodes, unlock moment, cues	FR-08, FR-09	10	10 passed	1.15
MapTapRulesTest	Map: tap rules	FR-08	4	4 passed	0.00
MapLayoutTest	Map: trail layout	NFR-ACC-02	2	2 passed	0.01
MapPathGeometryTest	Map: trail geometry	NFR-ACC-02	5	5 passed	0.01
MapTextFloorTest	Map: 16 sp text floor	NFR-ACC-02	1	1 passed	0.18
TopStatsBarLayoutTest	Map: header card, 64 dp closes	NFR-ACC-02	6	6 passed	1.43
SplashScreenTest	Splash: start, reduced motion	NFR-ACC-02, NFR-IND-01	4	4 passed	0.38
DimensTest	Design system: 64 dp, 16 sp floors	NFR-ACC-02	5	5 passed	0.00
GummyContainerLayoutTest	Design system: container sizing	NFR-ACC-02	3	3 passed	0.56
PlayItMotionTest	Design system: motion, reduced motion	NFR-ACC-02	2	2 passed	0.00
FeedbackEffectsTest	Design system: feedback effects	NFR-ACC-02	3	3 passed	0.00
IdleTimerTest	Idle re-prompt timer (10 s)	NFR-IND-01	6	6 passed	0.21
IdleWiggleTest	Idle wiggle on the main button	NFR-IND-01	4	4 passed	0.00
AssetImageTest	Pictures: loading	NFR-REL-01	4	4 passed	0.03
AssetImageLoadTest	Pictures: no stale picture	NFR-ACC-01	3	3 passed	10.11
ManifestPrivacyTest	Privacy: backup disabled	NFR-SEC-01	1	1 passed	0.03
ZeroEmojiPolicyTest	Policy: zero emoji	Team policy	1	1 passed	0.50
PerformancePolicyTest	Policy: no haptics, portrait, no endless idle motion	Team policy	4	4 passed	0.36
ViewModelInitOrderTest	Code health: init order	NFR-REL-01	1	1 passed	0.05
LayoutMatrixCompactTest	Layout matrix, 360x640	NFR-ACC-02	21	15 passed, 6 skipped	2.65

Test class	Feature	Requirement	Tests	Result	Time (s)
`LayoutMatrixA21sTest`	Layout matrix, 360x740 (A21s)	NFR-ACC-02	21	21 passed	7.27
`LayoutMatrixPhoneTest`	Layout matrix, 411x891	NFR-ACC-02	21	21 passed	3.57
`LayoutMatrixTabletTest`	Layout matrix, 800x1280	NFR-ACC-02	21	21 passed	1.99
Total (66 classes)			**433**	**427 passed, 6 skipped, 0 failed**	63.02

5.3 Skipped Tests

The 6 skips are the font-scale 1.3 checks on the 360x640 profile (`fontScale13_mainActionVisible`, `sayIt_micVisible_fontScale13`, `findIt_fontScale13`, `complete_continueVisible_fontScale13`, `splash_startVisible_fontScale13`, `nameprompt_letsPlayVisible_fontScale13`). They skip by design through `assumeFontScaleChecks()`: a 360x640 screen at 130% text is below the supported floor. The same 6 checks pass on the other three profiles (§6.2).

5.4 Inspections

APK permissions (TC-SEC-01). `tools/docs/inspect_apk.py` ran `aapt dump permissions` on the submitted file on October 18, 2026 (full log in `docs/std/evidence/2026-10-18-submitted-apk/apk_inspection.txt`); the run of October 14 on the earlier build of the same commit gave the same permissions (`docs/std/evidence/2026-10-14-0320385/apk_inspection.txt`). Output:

```
package: com.playit.app
uses-permission: name='android.permission.RECORD_AUDIO'
uses-permission: name='android.permission.READ_EXTERNAL_STORAGE' maxSdkVersion='32'
uses-permission: name='android.permission.WRITE_EXTERNAL_STORAGE' maxSdkVersion='28'
permission: com.playit.app.DYNAMIC_RECEIVER_NOT_EXPORTED_PERMISSION
uses-permission: name='com.playit.app.DYNAMIC_RECEIVER_NOT_EXPORTED_PERMISSION'
SHA-256 b5a54c18d1405c3851b7fec5c110f03f1aeaaa899a343b12f1796e04398c42d4
package com.playit.app, versionName 1.0.0, minSdk 26, targetSdk 34
```

Audio releases (TC-AUD-03). `tools/docs/check_audio_release.py` read the 7 release manifests in `docs/audio-release/` (76 clips) and compared each with the app's file by SHA-256: 69 identical, 0 different, 7 approved but not used by APK H. Of the 228 audio files the app ships, 69 come from a release and 159 are MVP clips from before the release gate (words 115, phonemes 29, UI lines 15); the SRS lists their remake and teacher audit as pending.

5.5 Device Tests on the Samsung Galaxy A21s

The team played each candidate build on the test phone as a child would. These sessions are exploratory, not scripted; what they found is in §7.1. The last row is the acceptance check of the submitted APK, run before turning it in.

Date	Build	What was checked	Result
October 10, 2026	APK B	One letter end to end; the Hear It sequence; captions	7 changes settled (card 28, `ac8e8c7`), among them: Hear It ends on the sound; Say It never traps the child; full Blend It sound-out; live star totals; one-tap map start; captions out of the child view (team decision)

Date	Build	What was checked	Result
October 11, 2026	APK D	Section 1 on the A21s	INC-01, INC-02, INC-03
October 11, 2026	APK E	Section 1 after card 28b	INC-04; the mouth cue moved to Say It only (card 28c)
October 18, 2026	Submitted APK (`b5a54c18`)	Acceptance: install on Samsung Galaxy A21s, open splash screen, verify map header, play Letter M through Hear It, Say It, Find It, and verify Letter Complete	PASS. Application installed cleanly, responsive audio playback, mic permission requested and handled, lesson progression verified on hardware.

6. Layout Verification

6.1 Method

`LayoutMatrixTest` renders the real screens with Robolectric on the 4 profiles of §2.5 and checks positions with unclipped bounds, so a control that is pushed below the window or behind another one fails. The map uses a 16-character learner name ("Maximilianooooooo"); the font-scale checks use 130% text. Each check also saves a capture (Roborazzi) for a visual review. 21 checks on 4 profiles make 84 runs: 78 passed and 6 skipped by design.

6.2 Results of the Layout Checks

Layout check	What it asserts	360x640	360x740	411x891	800x1280
`lessonScaffold_ bottomBarAndBodyVisible`	Lesson frame: body and bottom bar end inside the window	Pass	Pass	Pass	Pass
`fontScale13_mainActionVisible`	Lesson frame at font scale 1.3: same	Skip*	Pass	Pass	Pass
`tablet_contentWidthCapped`	Lesson content is at most 560 dp wide	Pass	Pass	Pass	Pass
`hearIt_playVisible`	Hear It: play button inside the window	Pass	Pass	Pass	Pass
`hearIt_stepsAboveNext`	Hear It: step dots end above Next	Pass	Pass	Pass	Pass
`sayIt_micVisible`	Say It: mic inside the window	Pass	Pass	Pass	Pass
`sayIt_micVisible_fontScale13`	Say It at font scale 1.3: mic inside the window	Skip*	Pass	Pass	Pass
`sayIt_mouthCueBesideMic`	Say It after two misses: mouth picture beside the mic, mic on screen	Pass	Pass	Pass	Pass
`findIt_wholeGridVisible`	Find It: whole picture grid inside the window	Pass	Pass	Pass	Pass
`findIt_fontScale13`	Find It at font scale 1.3: same	Skip*	Pass	Pass	Pass

Layout check	What it asserts	360x640	360x740	411x891	800x1280
`blendIt_tilesVisible`	Blend It: letter tiles inside the window	Pass	Pass	Pass	Pass
`map_currentNodeVisible`	Map (16-letter name): current letter node inside the window	Pass	Pass	Pass	Pass
`map_buttonsAre64dp`	Map: back and guidebook buttons at least 64 x 64 dp	Pass	Pass	Pass	Pass
`map_unitBannerTitleFits`	Map: unit title inside the banner face	Pass	Pass	Pass	Pass
`complete_continueVisible`	Letter complete: Continue inside the window	Pass	Pass	Pass	Pass
`complete_continueVisible_fontScale13`	Letter complete at font scale 1.3: same	Skip*	Pass	Pass	Pass
`splash_startVisible`	Splash: Start inside the window	Pass	Pass	Pass	Pass
`splash_startVisible_fontScale13`	Splash at font scale 1.3: Start inside, below Lily's face	Skip*	Pass	Pass	Pass
`splash_lilyFaceClearOfStart`	Splash: panel, paws and Start below Lily's face	Pass	Pass	Pass	Pass
`nameprompt_letsPlayVisible`	New learner: Let's play inside the window	Pass	Pass	Pass	Pass
`nameprompt_letsPlayVisible_fontScale13`	New learner at font scale 1.3: same	Skip*	Pass	Pass	Pass

* Skipped by design: 360x640 at font scale 1.3 is below the supported floor (§5.3).

6.3 Visual Review of the Captures

The figures put the same screen side by side on the 4 profiles at their true relative size (0.55 px per dp). They were captured on October 14, 2026 from the code of APK H.

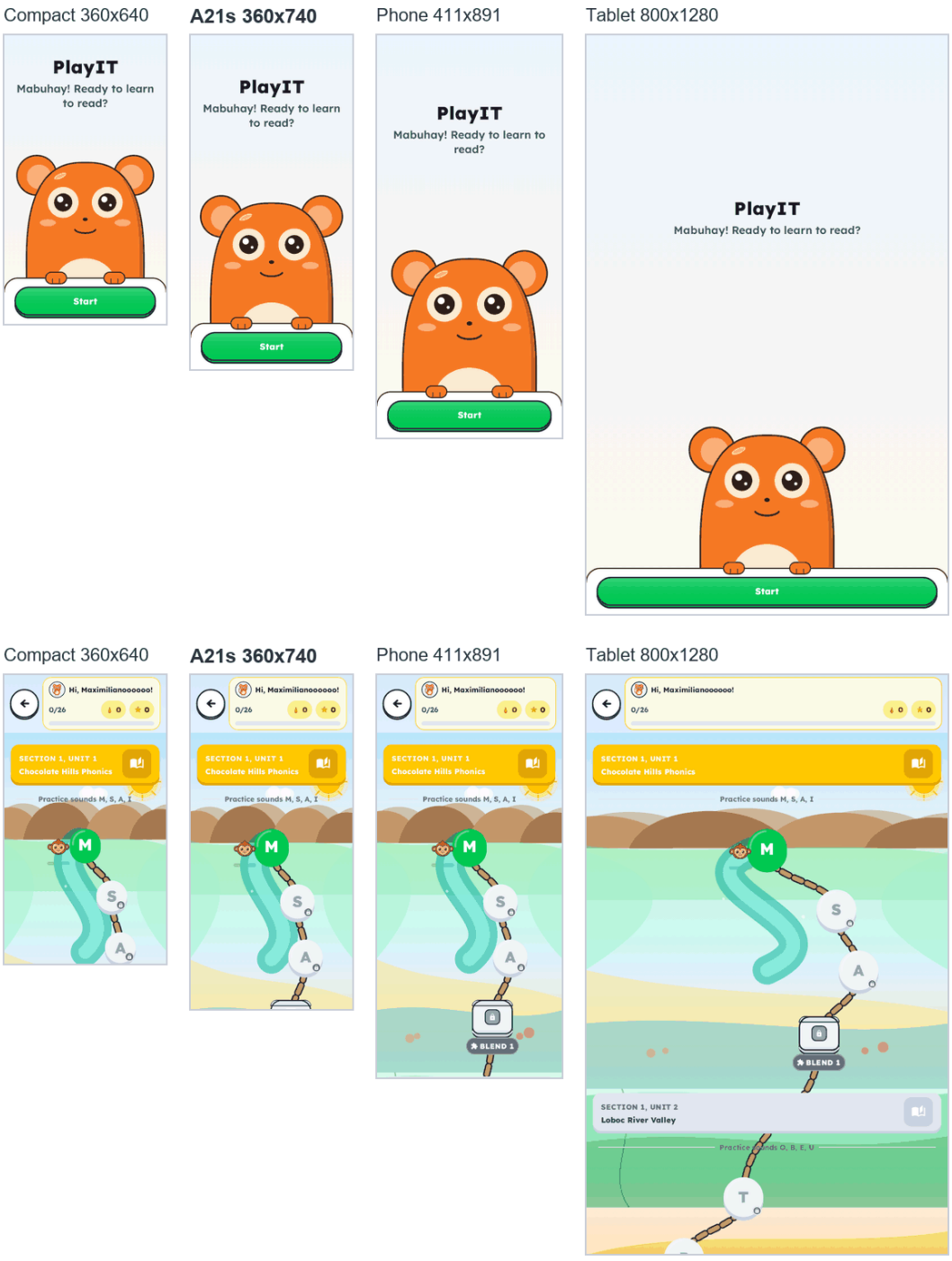


Figure 1. *Splash (top) and map (bottom) on the 4 profiles. Start stays below Lily's face; the map header holds a 16-character name.*

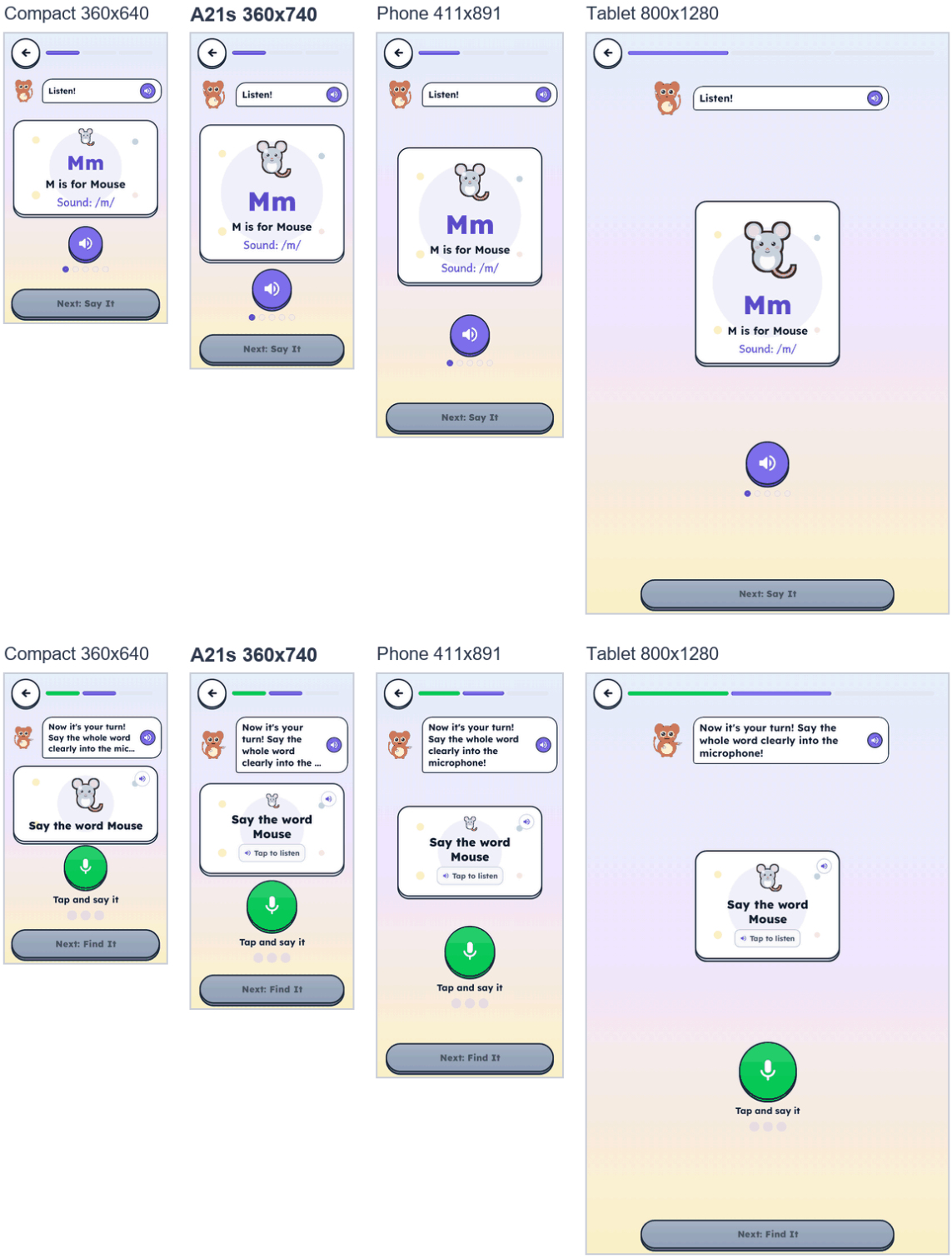


Figure 2. *Hear It while the sound plays (top) and Say It before an attempt (bottom).*

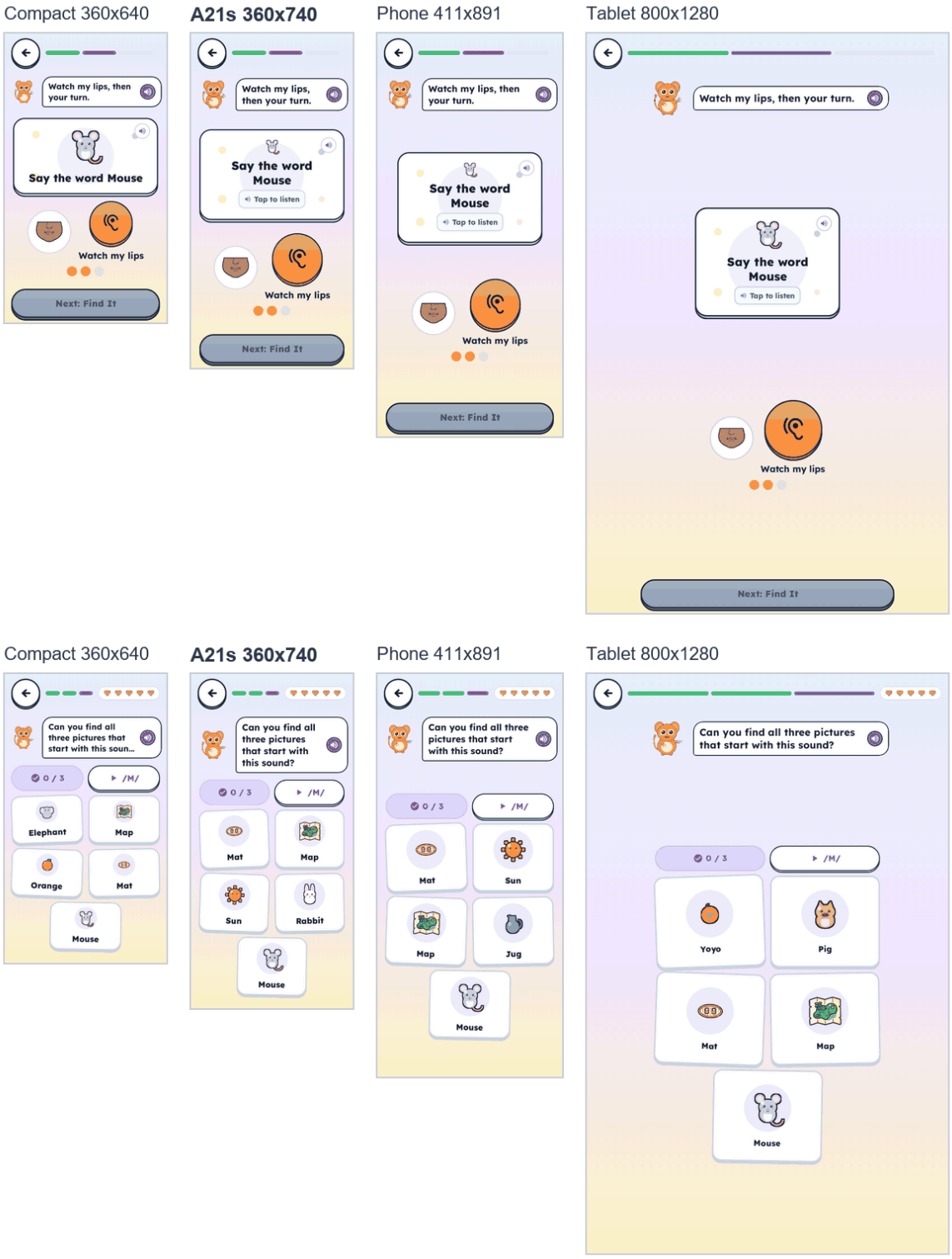


Figure 3. *Say It after two misses, with the mouth-shape cue beside the mic (top); Find It for the letter M (bottom).*

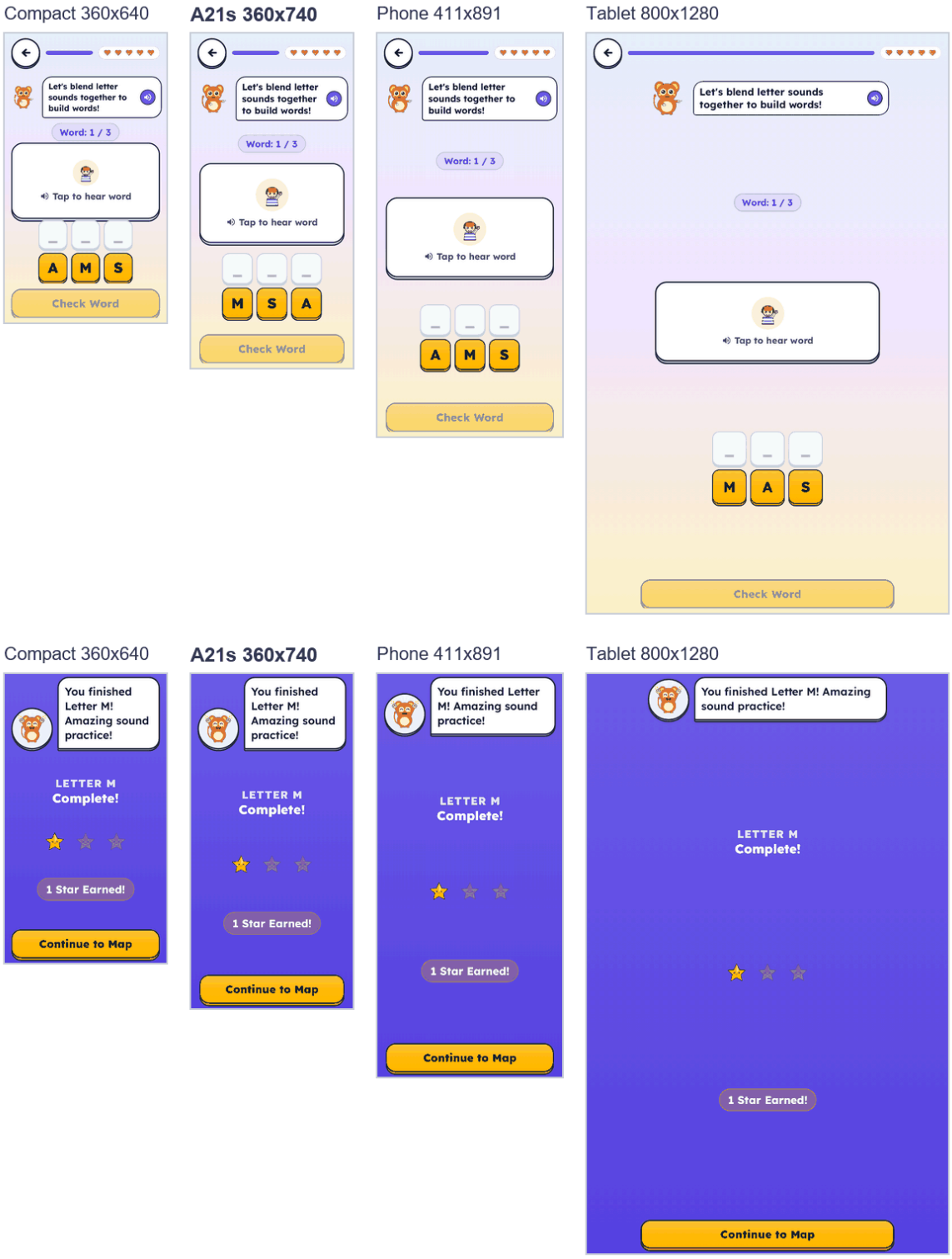


Figure 4. Blend It with the tiles dealt out of order (top) and Letter Complete with 1 star (bottom).

What the review found:

- Every control the child needs is on screen on all 4 profiles, which matches §6.2.

- **Figure 3, Find It for M:** the distractors are Elephant and Orange (360x640), Rabbit and Sun (360x740), Sun and Jug (411x891), Yoyo and Pig (800x1280). M is the first letter, so none of these sounds has been taught. This is DEF-01.
- **Figures 2 and 3, Lily's speech bubble:** the Say It line ("Now it's your turn! Say the whole word clearly into the microphone!") is cut with "…" on 360x640 and on the A21s profile, and the Find It line is cut on 360x640. The line is also spoken, so a pre-reader loses nothing, but the cut text looks unfinished (OBS-02).
- **Figure 4:** the Blend It tiles are dealt out of order on every profile (A-M-S, M-S-A, A-M-S, M-A-S for SAM), as TC-BLD-02 requires.

7. Incident and Defect Report

Severity: **High** — a child cannot finish a lesson or data is lost; **Medium** — a requirement is not met but the lesson works; **Low** — cosmetic or a gap in the tests.

7.1 Incidents Found and Closed

ID	Found (date, how, build)	Expected	Actual	Fix (commit)	Re-test evidence	Status
INC-01	Oct 11, phone test, APK D	Blend It plays the words AM, TUB and YAM	Silent: the clips were never released into the app	Audio release `2026-10-11-words` (ElevenLabs, after the Kokoro TUB takes failed review); card 15b, `b99d839`	TC-BLD-06; TC-AUD-03	Closed
INC-02	Oct 11, phone test, APK D	Each letter shows the mouth shape that makes its sound	Several letters showed the wrong mouth shape	Letter-to-mouth pairs remapped; card 28b, `ed997b6`	TC-ACC-01 (`ArticulationGroupTest`)	Closed
INC-03	Oct 11, phone test, APK D	Unit banner title and Say It attempt dots fit on small phones	Title clipped; dots overlapped on 360x640	Card 28b, `ed997b6`	`map_unitBannerTitleFits` on 4 sizes (§6.2)	Closed
INC-04	Oct 11, phone test, APK E	After "next letter" the card shows the new letter's picture	The mouse from M stayed on most letters of section 1	Picture loader restarts on a new path; card 28d, `e7f870e`	TC-ACC-03 (`AssetImageLoadTest`)	Closed
INC-05	Oct 1, code review	Stars and hearts follow the child's results	Stars were always 3; hearts lost before a restart vanished; Blend It at 0 hearts showed nothing; 2 of 5 hearts were not drawn	Card 11, `0d9ad3a`	TC-FND-06, TC-BLD-05, TC-PRG-01	Closed
INC-06	Oct 13, card 31e (correcting the card 30 audit, which had called the floor met)	Every child-facing control is at least 64 dp	Map back 46 dp, guidebook 44 dp, dialog closes 32 and 36 dp, dialog buttons 52–56 dp	Card 31e, `1420c84`	TC-ACC-02 (`map_buttonsAre64dp` on 4 sizes, `TopStatsBarLayoutTest`)	Closed
INC-07	Oct 13, UI audit (card 31)	Blend It never starts with the word already built	A random deal could match the answer	Card 31f, `0320385`	TC-BLD-02	Closed

7.2 Open Defects and Observations

ID	Req.	Description (expected vs actual)	Severity	Recommended action	Owner
DEF-01	FR-04	Expected: Find It distractors only from letters already mastered. Actual: drawn from every curriculum letter; `generate5ItemGrid` ignores the letters it is given. Letters 1 and 2 have no mastered letters, so the rule also needs the curated fallback that `01_REQUIREMENTS_SUMMARY.md` specified.	Medium	Draw distractors from the profile's completed letters; use a fixed fallback set for the first letters; add a test that a new profile's grid for S uses only M.	Developer (fix card)
DEF-02	FR-06	Expected (SRS): 2 stars need at least 80% accuracy. Actual: stars use hearts lost only, so 1 wrong tap (75%) gives 2 stars. With 3 targets the SRS rule cannot give 2 stars in Find It at all.	Low	Team decision: either amend FR-06 to the hearts-lost rule the app uses (0 → 3, 1–2 → 2, 3+ → 1, as `01_REQUIREMENTS_SUMMARY.md` §7.4 flagged), or pass accuracy to `StarCalculator`. Then align the test.	Team, with the adviser
OBS-01	FR-03.4	The SRS starts the prompt ladder after 3 s of silence; the app listens up to 3.8 s before judging. No test covers a fully silent attempt from end to end (TC-ASR-09 and TC-ASR-10 cover its parts).	Low	Amend FR-03.4 to 3.8 s, or shorten the timer; add a ViewModel test for silence.	Team
OBS-02	NFR-ACC-02	Lily's speech bubble is cut with "…" on 360 dp wide screens (Say It on 360x640 and 360x740; Find It on 360x640). The line is spoken, so nothing is lost for a pre-reader.	Low	Card 31c shortens both lines ("Your turn! Say the word."; "Find the pictures that start with this sound."); it waits for the team lead's listening review.	Team lead (review)
OBS-03	FR-04	`GridGeneratorTest.letterOne_usesFallbackDistractorsWhenNotEnoughMasteredLetters` tests `generateGrid`, which Find It no longer calls; it does not test the screen's grid.	Low	Retarget the test to `generate5ItemGrid` when DEF-01 is fixed, and remove the unused function.	Developer
OBS-04	NFR-SEC-01	The missing INTERNET permission is checked by inspection only (TC-SEC-01); no automated test fails if a library adds it.	Low	Add a test on the merged manifest.	Developer

8. Test Summary Report

8.1 Results in Numbers

- **Automated suite:** 433 test cases in 66 classes; **427 passed, 0 failed, 6 skipped by design** (§5).
- **Layout matrix:** 21 checks on 4 profiles = 84 runs; 78 passed, 6 skipped by design (§6).
- **Test cases specified in this STD:** 83.
  - **Pass:** 57, of which 2 by inspection of the build.

  - **Partial:** 1 (TC-DB-03).
  - **Fail:** 2 (TC-FND-03 = DEF-01, TC-PRG-02 = DEF-02).
  - **Not run:** 23 (§4.8: field measures, device timing and planned features).
- **Incidents:** 7 found in testing and closed (§7.1); 2 open defects and 4 observations (§7.2).

8.2 Assessment

APK H does what the SRS requires for every behaviour that can be checked on the build, with two exceptions found while writing this document: Find It's distractor source (DEF-01, Medium) and the accuracy term of the star rule (DEF-02, Low). Neither stops a child from finishing a lesson or loses data, so neither blocks the midterm submission. Both are stated here rather than left for the adviser to find.

The automated tests prove the logic and the layout, not the experience. The requirements that matter most for the research objectives still need people and the phone: teacher agreement with the speech judge (TC-ASR-01 to 06), the purity of the sounds (TC-AUD-02), latency (TC-PERF), and independent completion (TC-IND-03). They are scheduled in Round 2 (SPMP Milestones 5 and 6). Three items the SPMP placed in Milestone 4 (Week 6) are not in APK H: telemetry, the CSV export and the Gate 3 teacher audit. Their test cases (TC-TEL-01, TC-EXP-01, TC-EXP-02, TC-AUD-02) move with them, and the latency measures (TC-PERF) depend on the telemetry, so the schedule needs re-baselining before Round 2.

8.3 Next Steps

1. Write the fix card for DEF-01; take DEF-02 and OBS-01 to the team and the adviser as SRS amendments.
2. Add the tests named in OBS-03 and OBS-04.
3. Re-baseline the SPMP for the Milestone 4 items not yet built (telemetry, CSV export) and the Gate 3 teacher audit.
4. When cards 31b and 31c ship, re-run this suite and the layout matrix, and update this document (OBS-02 should close).
5. Round 2: run the test cases of §4.8 and record them in version 2.0 of this document (SPMP Milestone 7: "verified STD execution logs").

8.4 Sign-Off

Role	Name	Signature	Date
Prepared by (test execution and write-up)	Capstone Team		October 18, 2026
Reviewed by (Team Lead)	Riva, Zendrix	_____	October 18, 2026
Adviser	Mr. Joemarie C. Amparo	_____	

Appendix A. Test Execution Log

Every test case of the recorded run (§5.1), from the JUnit XML in `docs/std/evidence/2026-10-14-0320385/junit/`.

#	Test class	Test case	Result	Time (s)
1	`HearItSequenceBuilderTest`	`build_m_mouse_followsTable3`	Pass	0.004

#	Test class	Test case	Result	Time (s)
2	HearItSequenceBuilderTest	build_noKeyWord_dropsKeyword	Pass	0.000
3	HearItSequenceBuilderTest	pauseMillis_parsesOnlyPauseTokens	Pass	0.000
4	HearItSequenceBuilderTest	replayTemplate_isSteps3to6	Pass	0.002
5	HearItViewModelTest	caption_followsSequence	Pass	0.066
6	HearItViewModelTest	firstSequenceEnd_playsNextCue_andHighlights	Pass	0.062
7	HearItViewModelTest	idle_afterHighlight_playsNextCue	Pass	0.064
8	HearItViewModelTest	idle_nudgesTheScreen	Pass	0.044
9	HearItViewModelTest	loadPhoneme_invalidId_setsLoadError	Pass	0.042
10	HearItViewModelTest	loadPhoneme_nullId_defaultsToId1	Pass	0.071
11	HearItViewModelTest	loadPhoneme_validId_loadsPhoneme	Pass	0.046
12	HearItViewModelTest	load_pendingExampleWord_dropsKeyword	Pass	0.091
13	HearItViewModelTest	load_playsFullModelingSequence	Pass	0.109
14	HearItViewModelTest	nextStaysOff_whileFirstSequencePlays	Pass	0.090
15	HearItViewModelTest	noIdlePrompt_withoutScreenVisible	Pass	0.056
16	HearItViewModelTest	playPhonemeSound_incrementsPlayCount	Pass	0.450
17	HearItViewModelTest	playPhonemeSound_playsReplaySegment	Pass	0.082
18	HearItViewModelTest	replayEnd_alsoUnlocks	Pass	0.095
19	HearItScreenFocusTest	whilePlaying_playButtonKeepsItsColour	Pass	2.587
20	HearItScreenFocusTest	whilePlaying_soundTextShowsOnce_ andLilySaysListen	Pass	0.406
21	HearItScreenshotTest	hearIt_letterM	Pass	0.270
22	AudioResolverTest	correctionFragments_fileExists	Pass	0.002
23	AudioResolverTest	devPlaceholder_mapsKeywordsAndTutor	Pass	0.001
24	AudioResolverTest	devPlaceholder_mapsUiLines	Pass	0.002
25	AudioResolverTest	getDevPlaceholderForAsset_ mapsProductionPathsToDevPlaceholders	Pass	0.000
26	AudioResolverTest	getDevPlaceholderPath_ returnsCorrectCategoryPaths	Pass	0.000
27	AudioResolverTest	getKeyWordPath_returnsWavInKeywordsFolder	Pass	0.008
28	AudioResolverTest	getPhonemePath_ returnsCorrectPathForSpecialLetters	Pass	0.000
29	AudioResolverTest	getPhonemePath_ returnsCorrectPathForStandardLetters	Pass	0.000
30	AudioResolverTest	getPhonemePath_returnsDraftPathForX	Pass	0.002
31	AudioResolverTest	getSfxPath_returnsCorrectPaths	Pass	0.002
32	AudioResolverTest	getTutorPath_returnsWavInTutorFolder	Pass	0.001
33	AudioResolverTest	getUiPath_returnsWavInUiFolder	Pass	0.000

#	Test class	Test case	Result	Time (s)
34	AudioResolverTest	getVoPath_returnsCorrectPaths	Pass	0.001
35	AudioResolverTest	getWordPath_returnsCorrectPath	Pass	0.000
36	AudioResolverTest	lessonVo_allReleasedLinesUseWav	Pass	0.026
37	AudioResolverTest	otherPhonemes_keepOldPath	Pass	0.000
38	AudioResolverTest	phonemeI_usesReleasedClip	Pass	0.001
39	AudioResolverTest	phonemeM_usesReleasedHeldSound	Pass	0.001
40	AudioResolverTest	phonemeS_usesReleasedHeldSound	Pass	0.001
41	AudioResolverTest	releasedPhoneme_fileExists	Pass	0.000
42	AudioResolverTest	releasedWords_filesExist	Pass	0.001
43	AudioResolverTest	releasedWords_useWav	Pass	0.001
44	AudioResolverTest	rotationalHelpers_alternateCorrectly	Pass	0.004
45	AudioCompletenessCheckTest	approvedTutorCarriers_exist	Pass	0.003
46	AudioCompletenessCheckTest	keywordClips_existForAll26SeededWords	Pass	0.012
47	AudioCompletenessCheckTest	verifyAll28PhonemesArePresent	Pass	0.001
48	AudioCompletenessCheckTest	verifyAllRequiredAudioFilesExistOnDisk	Pass	0.008
49	AudioCompletenessCheckTest	verifyAllSayItWordPromptsExistOnDisk	Pass	0.012
50	AudioCompletenessCheckTest	verifyRequiredAssetCounts	Pass	0.001
51	CaptionTextTest	carrier	Pass	0.000
52	CaptionTextTest	keywordClip	Pass	0.002
53	CaptionTextTest	pauseHasNoCaption	Pass	0.000
54	CaptionTextTest	phonemeClip	Pass	0.001
55	SayItViewModelTest	addedVowelError_ triggersFbNoAhSpokenCorrection	Pass	0.053
56	SayItViewModelTest	afterLeadAndMoveOn_startListeningIsIgnored	Pass	0.046
57	SayItViewModelTest	correctAfterMiss_praises_andCanContinue	Pass	0.067
58	SayItViewModelTest	evaluateSpeech_addedVowel_ setsIncorrectWithAddedVowel	Pass	0.054
59	SayItViewModelTest	evaluateSpeech_correctTranscript_ setsCorrectState	Pass	0.062
60	SayItViewModelTest	evaluateSpeech_incorrectTranscript_ doesNotDeductHeart	Pass	0.057
61	SayItViewModelTest	evaluateSpeech_incorrectTranscript_ setsIncorrectState	Pass	0.034
62	SayItViewModelTest	evaluateSpeech_letterName_ setsIncorrectWithLetterName	Pass	0.041
63	SayItViewModelTest	evaluateSpeech_ recordsMultipleAttemptsInList	Pass	0.090
64	SayItViewModelTest	evaluateSpeech_setsLastHeard	Pass	0.042

#	Test class	Test case	Result	Time (s)
65	SayItViewModelTest	firstLetterNameMiss_ playsLetterNameCorrection	Pass	0.079
66	SayItViewModelTest	idle_noNudgeOnceHidden	Pass	0.061
67	SayItViewModelTest	idle_tenSecondsWithoutTouch_nudgesOnce	Pass	0.056
68	SayItViewModelTest	idle_waitsWhileTheModelPlays	Pass	0.028
69	SayItViewModelTest	lastHeard_tracksAttemptNumber	Pass	0.028
70	SayItViewModelTest	letterSoundMode_correctionDropsKeyWord	Pass	0.061
71	SayItViewModelTest	loadPhoneme_invalidId_setsLoadError	Pass	0.031
72	SayItViewModelTest	loadPhoneme_resetsLastHeard	Pass	0.034
73	SayItViewModelTest	loadPhoneme_smePendingLetter_ usesLegacyLetterMode	Pass	0.057
74	SayItViewModelTest	loadPhoneme_validId_loadsPhoneme	Pass	0.041
75	SayItViewModelTest	loadPhoneme_wordMode_ exposesNormalizedTargetWord	Pass	0.042
76	SayItViewModelTest	onResult_finalLetterName_ setsIncorrectWithLetterName	Pass	0.106
77	SayItViewModelTest	onResult_partialAddedVowelThenFinalWord_ isCorrect	Pass	0.115
78	SayItViewModelTest	onResult_partialAddedVowelThenTimeout_ setsIncorrectWithAddedVowel	Pass	0.071
79	SayItViewModelTest	onResult_partialFoil_keepsListening	Pass	0.078
80	SayItViewModelTest	onResult_partialLetterNameThenFinalWord_ isCorrect	Pass	0.090
81	SayItViewModelTest	otherWordMiss_playsListenRemodel	Pass	0.046
82	SayItViewModelTest	partialSpeech_setsHeard	Pass	0.065
83	SayItViewModelTest	playWordAudio_rapidTaps_debounced	Pass	0.081
84	SayItViewModelTest	playWordAudio_whenListening_ stopsListeningAndPlaysAudio	Pass	0.088
85	SayItViewModelTest	recognizerStoppedExternally_returnsToIdle	Pass	0.057
86	SayItViewModelTest	secondMiss_playsWatchMyLips	Pass	0.049
87	SayItViewModelTest	secondMiss_showsMouthCue	Pass	0.622
88	SayItViewModelTest	startListening_setsLetterScopedGrammar	Pass	0.052
89	SayItViewModelTest	thirdMiss_emitsLeadAndMoveOn_ andSavesIncorrect	Pass	0.058
90	SayItViewModelTest	thirdMiss_micRestsAndNextUnlocks	Pass	0.085
91	SayItViewModelTest	wordMode_letterSoundTranscript_isIncorrect	Pass	0.060
92	SayItViewModelTest	wordMode_promptPlaysIntroVoThenWordAudio	Pass	0.080
93	SayItFeedbackCopyTest	addedVowel_level1_namesTheError	Pass	0.001
94	SayItFeedbackCopyTest	leadAndMoveOn_neverSaysTryAgain	Pass	0.001

#	Test class	Test case	Result	Time (s)
95	SayItFeedbackCopyTest	letterName_level1_namesTheError	Pass	0.000
96	SayItFeedbackCopyTest	level2_isWatchMyLips_forAnyType	Pass	0.004
97	SayItFeedbackCopyTest	noRowSaysTryAgain	Pass	0.009
98	SayItFeedbackCopyTest	noSpeech_soundMode_saysIt	Pass	0.002
99	SayItFeedbackCopyTest	noSpeech_wordMode_saysTheWord	Pass	0.001
100	SayItFeedbackCopyTest	praise_isEffortPraise	Pass	0.000
101	MicStatusTest	correct	Pass	0.000
102	MicStatusTest	idle	Pass	0.000
103	MicStatusTest	incorrect	Pass	0.003
104	MicStatusTest	listeningHeard	Pass	0.000
105	MicStatusTest	listeningSilent	Pass	0.000
106	MicStatusTest	movedOn_isDone	Pass	0.000
107	TutorPolicyTest	correct_anyAttempt_praises	Pass	0.000
108	TutorPolicyTest	firstMiss_correctsAtLevel1_withErrorType	Pass	0.000
109	TutorPolicyTest	secondMiss_correctsAtLevel2	Pass	0.000
110	TutorPolicyTest	thirdMiss_leadsAndMovesOn	Pass	0.000
111	SpeechValidatorTest	distractorRejection_returnsFalse	Pass	0.000
112	SpeechValidatorTest	grammarFor_soundMode_hasNoAcceptedTokens	Pass	0.000
113	SpeechValidatorTest	grammarFor_wordMode_ hasFoilsNotGenericDecoys	Pass	0.002
114	SpeechValidatorTest	judgeSound_addedVowel_rejected	Pass	0.006
115	SpeechValidatorTest	judgeSound_anchorWord_isOtherWord	Pass	0.000
116	SpeechValidatorTest	judgeSound_letterName_rejected	Pass	0.002
117	SpeechValidatorTest	judgeSound_noDuration_unconfirmed	Pass	0.001
118	SpeechValidatorTest	judgeSound_soundModeOff_ sustainedIsUnconfirmed	Pass	0.001
119	SpeechValidatorTest	judgeSound_soundModeOn_sustainedContinuous_ accepted	Pass	0.000
120	SpeechValidatorTest	judgeSound_soundModeOn_tooShort_rejected	Pass	0.002
121	SpeechValidatorTest	judgeSound_stop_neverAcceptedBySustain	Pass	0.001
122	SpeechValidatorTest	judgeSound_substitution_rejected	Pass	0.001
123	SpeechValidatorTest	judgeWord_paths	Pass	0.001
124	SpeechValidatorTest	legacyNgEnye_exactMatch_returnsTrue	Pass	0.000
125	SpeechValidatorTest	nullOrBlankText_returnsFalse	Pass	0.003
126	SpeechValidatorTest	phonemeMode_rejectsMisspelledAnchorWords_ afterCB1	Pass	0.000

#	Test class	Test case	Result	Time (s)
127	SpeechValidatorTest	soundMode_isOffByDefault	Pass	0.000
128	SpeechValidatorTest	validate_prefixRuleRemoved	Pass	0.000
129	SpeechValidatorTest	wordAcceptedVariants_ hasEntryForEverySeededExampleWord	Pass	0.002
130	SpeechValidatorTest	wordExactMatch_returnsTrue_ forAll26ExampleWords	Pass	0.000
131	SpeechValidatorTest	wordMode_caseAndWhitespaceInsensitive	Pass	0.000
132	SpeechValidatorTest	wordMode_hyphenatedYoyoVariant_returnsTrue	Pass	0.001
133	SpeechValidatorTest	wordMode_nullOrBlankTranscript_returnsFalse	Pass	0.003
134	SpeechValidatorTest	wordMode_rejectsLetterSoundOnly	Pass	0.000
135	SpeechValidatorTest	wordMode_rejectsOtherWordsAndMinimalPairs	Pass	0.000
136	SpeechValidatorTest	wordMode_rejectsPartialOrMisspelledWords	Pass	0.001
137	SpeechValidatorTest	wordMode_unknownTargetWord_ fallsBackToCanonicalOnly	Pass	0.000
138	ArticulationGroupTest	caseAndSpacesIgnored	Pass	0.001
139	ArticulationGroupTest	everyLetterHasAGroup	Pass	0.004
140	ArticulationGroupTest	everyLetterMapsToItsMouthShape	Pass	0.002
141	ArticulationGroupTest	jAndQRoundTheLips	Pass	0.000
142	ArticulationGroupTest	lipsTogether	Pass	0.001
143	ArticulationGroupTest	rDoesNotRoundTheLips	Pass	0.000
144	ArticulationGroupTest	shortVowelsAreNotRoundLips	Pass	0.000
145	ArticulationGroupTest	teethOnLip	Pass	0.000
146	ArticulationCueTest	missingAsset_rendersNothing	Pass	0.442
147	ArticulationCueTest	releasedPicture_renders	Pass	5.044
148	FindItViewModelTest	completion_appendsNextCue	Pass	0.048
149	FindItViewModelTest	idle_replaysIntro	Pass	0.074
150	FindItViewModelTest	loadGrid_invalidId_setsLoadError	Pass	0.041
151	FindItViewModelTest	loadGrid_validId_loadsTargetAnd5ItemGrid	Pass	0.020
152	FindItViewModelTest	restartSession_resetsHeartsAndState	Pass	0.022
153	FindItViewModelTest	selectPictureItem_allHeartsLost_ setsGameOver	Pass	0.371
154	FindItViewModelTest	selectPictureItem_allThreeCorrect_ completesLesson	Pass	0.028
155	FindItViewModelTest	selectPictureItem_correctItem_ incrementsFoundCount	Pass	0.045
156	FindItViewModelTest	selectPictureItem_incorrectItem_ deductsHeart	Pass	0.063
157	FindItViewModelTest	sessionHeartsLost_survivesRestart	Pass	0.028

#	Test class	Test case	Result	Time (s)
158	FindItViewModelTest	threeCorrectInARow_recoversAHeart	Pass	0.033
159	FindItViewModelTest	wrongTaps_countIntoSessionHeartsLost	Pass	0.029
160	GridGeneratorTest	distractorLettersFor_k_ excludesSameSoundGroup	Pass	0.000
161	GridGeneratorTest	distractors_neverFromSpecialBanks	Pass	0.147
162	GridGeneratorTest	distractors_neverShareTheTargetSound	Pass	0.203
163	GridGeneratorTest	generate5ItemGrid_ returnsExactly5ItemsWith3CorrectAnd2Distractors	Pass	0.001
164	GridGeneratorTest	letterOne_ usesFallbackDistractorsWhenNotEnoughMasteredLetters	Pass	0.001
165	GridGeneratorTest	standardGrid_containsTargetAndDistractors	Pass	0.000
166	GridGeneratorTest	up_usesItsOwnPicture	Pass	0.002
167	HeartManagerTest	checkRecovery_ addsHeartEveryThreeConsecutiveCorrect	Pass	0.000
168	HeartManagerTest	deductHeart_ decrementsCountAndIncrementsHeartsLost	Pass	0.000
169	HeartManagerTest	deductHeart_signalsGameOverWhenZero	Pass	0.000
170	HeartManagerTest	recoveryCapsAtStartingPool_recommendedRule	Pass	0.000
171	HeartManagerTest	reset_clearsSessionHeartsLost	Pass	0.001
172	HeartManagerTest	reset_restoresInitialStartingPool	Pass	0.000
173	HeartManagerTest	restartAtReducedPoolOnDepletion	Pass	0.000
174	HeartManagerTest	sessionHeartsLost_survivesRestart	Pass	0.000
175	HeartManagerTest	startsAtStartingPool	Pass	0.000
176	FindItScreenshotTest	findIt_letterM_fiveHearts	Pass	0.384
177	PictureAssetsTest	everyGridPictureExists	Pass	0.159
178	PictureAssetsTest	everyMouthCueIsReleased	Pass	0.028
179	PictureAssetsTest	releasedPicturesMatchManifest	Pass	0.259
180	BlendItViewModelTest	applyHint_afterTwoMistakes_ locksCorrectCharacter	Pass	0.021
181	BlendItViewModelTest	heartDepleted_restartsWithThreeHearts	Pass	0.024
182	BlendItViewModelTest	idle_replaysIntro	Pass	0.074
183	BlendItViewModelTest	loadSessionWords_ initializesFirstWordAndTileBank	Pass	0.843
184	BlendItViewModelTest	placeAndRemoveTiles_ updatesBankAndPlacedLists	Pass	0.022
185	BlendItViewModelTest	result_countsFirstTryWords	Pass	0.031
186	BlendItViewModelTest	soundOut_lightsAllTilesBeforeWord	Pass	0.103

#	Test class	Test case	Result	Time (s)
187	BlendItViewModelTest	soundOut_waitsForEachSound	Pass	0.096
188	BlendItViewModelTest	submitWord_correctWord_ transitionsToWordCorrectAndAdvances	Pass	0.047
189	BlendItViewModelTest	submitWord_incorrectWord_deductsHeart	Pass	0.057
190	BlendItViewModelTest	wrongSubmit_playsSoftPop_notBuzz	Pass	0.058
191	BlendItTileDealTest	deal_isNeverTheAnswer_ andAlwaysTheSameLetters	Pass	0.027
192	BlendItTileDealTest	deal_ofOneRepeatedLetter_keepsItsOnlyOrder	Pass	0.004
193	BlendItWordSeedsTest	everyWordUsesOnlyTaughtLetters	Pass	0.005
194	BlendItWordSeedsTest	nonDecodableWordsAreGone	Pass	0.000
195	BlendItWordSeedsTest	releasedWordsPointAtTheirWav	Pass	0.000
196	BlendItWordSeedsTest	replacementsKeepTheirSlots	Pass	0.000
197	BlendItWordSeedsTest	thirtyThreeWords_threeThenFivePerGroup	Pass	0.009
198	BlendItWordSelectorTest	groupOne_restrictedToThreeWords	Pass	0.000
199	BlendItWordSelectorTest	standardGroup_takesUpToFiveWords	Pass	0.001
200	BlendItCompleteViewModelTest	completeSession_ calculatesStarsAndSavesProgress	Pass	4.350
201	BlendItCompleteViewModelTest	completion_appendsNextCue	Pass	0.189
202	BlendItCompleteViewModelTest	idle_waitsForCompletionSequence	Pass	0.139
203	BlendItCompleteViewModelTest	starsUseNavResults	Pass	0.200
204	BlendItStarThresholdsTest	oneOrTwoHeartsLost_returnsTwoStars	Pass	0.001
205	BlendItStarThresholdsTest	threeOrMoreHeartsLost_returnsOneStar	Pass	0.000
206	BlendItStarThresholdsTest	zeroHeartsLost_returnsThreeStars	Pass	0.000
207	BlendItCardLayoutTest	hintLine_staysInsideCard	Pass	0.949
208	BlendItScreenshotTest	blendIt_group1_firstWord	Pass	0.574
209	StarCalculatorTest	oneOrTwoHeartsLost_returnsTwoStars	Pass	0.001
210	StarCalculatorTest	threeOrMoreHeartsLost_returnsOneStar	Pass	0.003
211	StarCalculatorTest	zeroHeartsLost_returnsThreeStars	Pass	0.001
212	LetterCompleteViewModelTest	completeLesson_ calculates2StarsFor1HeartLost	Pass	0.070
213	LetterCompleteViewModelTest	completeLesson_ calculates2StarsFor2HeartsLost	Pass	0.285
214	LetterCompleteViewModelTest	completeLesson_ calculates3StarsFor0HeartsLost_ andSavesProgress	Pass	0.105
215	LetterCompleteViewModelTest	completeLesson_handlesMissingPhoneme_ andSetsLoadError	Pass	0.063
216	LetterCompleteViewModelTest	completion_appendsNextCue	Pass	0.047

#	Test class	Test case	Result	Time (s)
217	LetterCompleteViewModelTest	heartsLostArg_missing_givesThreeStars	Pass	0.103
218	LetterCompleteViewModelTest	heartsLostArg_threeOrMore_givesOneStar	Pass	0.097
219	LetterCompleteViewModelTest	idle_waitsForCompletionSequence	Pass	0.038
220	LetterCompleteViewModelTest	retry_reloadsPhonemeSuccessfully	Pass	0.084
221	LetterCompleteMessageTest	oneStar_saysFinished	Pass	0.000
222	LetterCompleteMessageTest	twoOrThreeStars_sayMastered	Pass	0.003
223	LetterCompleteScreenshotTest	letterComplete_threeHeartsLost_oneStar	Pass	0.187
224	RoutesTest	blendItComplete_carriesResults	Pass	0.002
225	RoutesTest	blendItComplete_defaultsAreReasonable	Pass	0.001
226	RoutesTest	letterComplete_carriesHeartsLost	Pass	0.002
227	RoutesTest	letterComplete_defaultHeartsLostIsZero	Pass	0.000
228	UnlockManagerTest	phonemeN_lockedWhenPreviousNotCompleted	Pass	0.001
229	UnlockManagerTest	phonemeN_unlockedWhenPreviousCompleted	Pass	0.000
230	UnlockManagerTest	phonemeOne_unlockedByDefault	Pass	0.000
231	GroupUnlockManagerTest	groupN_ unlockedOnlyWhenAllPreviousMembersCompleted	Pass	0.003
232	GroupUnlockManagerTest	groupOne_unlockedByDefault	Pass	0.000
233	GroupUnlockManagerTest	isBlendItUnlocked_ lockedWhenGroupLettersNotCompleted	Pass	0.000
234	GroupUnlockManagerTest	isBlendItUnlocked_ unlockedWhenAllGroupLettersCompleted	Pass	0.000
235	StreakTrackerTest	checkMilestone_ returnsFormattedTitleForMilestonesOnly	Pass	0.001
236	StreakTrackerTest	consecutiveDayActivity_incrementsStreak	Pass	0.003
237	StreakTrackerTest	firstActivity_setsStreakToOne	Pass	0.002
238	StreakTrackerTest	inactivityGap_resetsStreakToOne_onActivity	Pass	0.003
239	StreakTrackerTest	milestoneBadges_unlockedAt5_10_15_20	Pass	0.004
240	StreakTrackerTest	resetIfInactive_resetsStreakToZero_ withoutAffectingStars	Pass	0.110
241	StreakTrackerTest	sameDayActivity_retainsCurrentStreak	Pass	0.004
242	LetterStatusCalculatorTest	3 or more failed attempts returns RED even with high accuracy	Pass	0.000
243	LetterStatusCalculatorTest	accuracy 80 percent or higher with low failed attempts returns GREEN	Pass	0.001
244	LetterStatusCalculatorTest	accuracy between 50 and 79 percent with low failed attempts returns YELLOW	Pass	0.000
245	LetterStatusCalculatorTest	accuracy under 50 percent returns RED	Pass	0.000
246	RetentionCalculatorTest	calculates percentage correctly for attempts within 7 days	Pass	0.003

#	Test class	Test case	Result	Time (s)
247	RetentionCalculatorTest	no attempts in 7-day window returns 0 retention	Pass	0.001
248	ParentDashboardViewModelTest	deleteActiveProfile_clearsSession	Pass	0.041
249	ParentDashboardViewModelTest	deleteLastProfile_setsNoProfilesLeft	Pass	0.032
250	ParentDashboardViewModelTest	deleteOtherProfile_keepsSession	Pass	0.041
251	ParentDashboardViewModelTest	deleteProfile_callsRepository	Pass	0.029
252	ParentDashboardViewModelTest	exportPdfReport_failure_updatesExportStatus	Pass	0.071
253	ParentDashboardViewModelTest	exportPdfReport_success_updatesExportStatus	Pass	0.169
254	ParentDashboardViewModelTest	init_ selectsFirstProfileAndLoadsDashboardData	Pass	0.027
255	ParentDashboardViewModelTest	init_withEmptyProfiles_setsLoadingFalse	Pass	0.015
256	ParentDashboardViewModelTest	renameProfile_blankOrTooLong_isIgnored	Pass	0.027
257	ParentDashboardViewModelTest	renameProfile_updatesTrimmedName	Pass	0.226
258	ParentDashboardViewModelTest	rename_refreshesSelectedProfile	Pass	0.029
259	ParentDashboardViewModelTest	selectProfile_handlesDashboardDataError	Pass	0.087
260	ParentDashboardViewModelTest	selectProfile_loadsDataForSelectedProfile	Pass	0.041
261	ArithmeticGateManagerTest	generateProblem_producesValidProblem_ andExpectedAnswerValidatesCorrectly	Pass	0.002
262	ArithmeticGateManagerTest	incorrectAnswer_isRejected	Pass	0.000
263	ArithmeticGateManagerTest	nonNumericOrEmptyString_isRejected	Pass	0.001
264	ArithmeticGateManagerTest	validateAnswer_ trimsLeadingAndTrailingWhitespace	Pass	0.001
265	VoiceCreditTest	creditNamesElevenLabsAndItsDomain	Pass	0.000
266	ProfileViewModelTest	arithmeticGuard_showAndDismissTransitions	Pass	0.002
267	ProfileViewModelTest	canAddProfile_returnsTrueWhenUnderLimit_ andFalseWhenAtMax	Pass	0.047
268	ProfileViewModelTest	createProfile_blankName_usesAvatarName	Pass	0.096
269	ProfileViewModelTest	createProfile_failure_emitsError	Pass	0.053
270	ProfileViewModelTest	createProfile_success_ updatesSessionAndEmitsCreated	Pass	0.056
271	ProfileViewModelTest	createProfile_typedName_isKept	Pass	0.064
272	ProfileViewModelTest	namePromptIntro_playsPickAvatar	Pass	0.143
273	ProfileViewModelTest	onAvatarSelected_ updatesAvatarIdAndPlaysSound	Pass	0.022
274	ProfileViewModelTest	resetForm_restoresDefaultValues	Pass	0.006
275	ProfileViewModelTest	selectProfile_ stopsAudioAndSetsActiveProfile	Pass	0.036
276	ProfileDaoStarsTest	getAllProfiles_reEmitsWhenProgressChanges	Pass	0.246

#	Test class	Test case	Result	Time (s)
277	ProfileDaoStarsTest	totalStars_isPerProfile	Pass	0.262
278	ProfileDaoStarsTest	totalStars_isZeroWithoutProgress	Pass	0.176
279	ProfileDaoStarsTest	totalStars_sumsLessonAndBlendStars	Pass	6.112
280	NamePromptScreenshotTest	namePrompt_avatarOnly	Pass	0.128
281	MapViewModelTest	audioActions_invokeAudioPlayerCorrectly	Pass	0.040
282	MapViewModelTest	clearSession_clearsActiveProfile	Pass	0.021
283	MapViewModelTest	init_ recordsActivityAndResetsStreakIfInactive	Pass	0.040
284	MapViewModelTest	lockedNode_playsLockedCue	Pass	0.017
285	MapViewModelTest	mapNodes_generatesLetterNodesAndBlendItNode	Pass	0.088
286	MapViewModelTest	onUnlockShown_clears	Pass	0.775
287	MapViewModelTest	returnFromLesson_afterResubscribe_ setsNewlyUnlocked	Pass	0.052
288	MapViewModelTest	unlockBetweenLoads_setsNewlyUnlocked	Pass	0.053
289	MapViewModelTest	unlockedNode_playsStartCue	Pass	0.024
290	MapViewModelTest	userStats_aggregatesProfileAndAchievements	Pass	0.035
291	MapTapRulesTest	activeNode_launchesDirectly	Pass	0.001
292	MapTapRulesTest	greeting_fitsOneLine	Pass	0.000
293	MapTapRulesTest	lockedNode_staysLocked	Pass	0.000
294	MapTapRulesTest	otherUnlockedNode_opensPopup	Pass	0.001
295	MapLayoutTest	offsetsStayInsideTheWidth	Pass	0.003
296	MapLayoutTest	widerScreen_widerPath	Pass	0.002
297	MapPathGeometryTest	calculateNodeXOffsetDp_isDeterministic	Pass	0.000
298	MapPathGeometryTest	calculateNodeXOffsetDp_ oscillatesLeftAndRight	Pass	0.000
299	MapPathGeometryTest	generateCompanionPlacements_ placesExplorerLeaderAndFriends	Pass	0.004
300	MapPathGeometryTest	generateTerrainProps_ createsSeededPropsForNodes	Pass	0.005
301	MapPathGeometryTest	getMarungkoLettersForGroup_ returnsExpectedLetters	Pass	0.001
302	MapTextFloorTest	noMapTextUnder16sp	Pass	0.179
303	TopStatsBarLayoutTest	card_isLilysTapTarget	Pass	0.219
304	TopStatsBarLayoutTest	guidebookDialog_closeIs64dp	Pass	0.581
305	TopStatsBarLayoutTest	lettersCount_isNotCut	Pass	0.095
306	TopStatsBarLayoutTest	longName_greetingIsNotCut	Pass	0.160
307	TopStatsBarLayoutTest	longName_pillsStayVisible	Pass	0.176

#	Test class	Test case	Result	Time (s)
308	TopStatsBarLayoutTest	nodeDialog_closeIs64dp	Pass	0.193
309	SplashScreenTest	blink_closesBothEyes	Pass	0.076
310	SplashScreenTest	motionPolicy_followsReducedMotion	Pass	0.070
311	SplashScreenTest	start_callsOnStart	Pass	0.147
312	SplashScreenTest	wink_closesOnlyTheRightEye_thenReopens	Pass	0.089
313	DimensTest	a21sIsRegular	Pass	0.000
314	DimensTest	childTargetsNeverBelow64dp	Pass	0.000
315	DimensTest	compactUnder700dpTall	Pass	0.000
316	DimensTest	tabletIsWide	Pass	0.000
317	DimensTest	textNeverBelow16sp	Pass	0.004
318	GummyContainerLayoutTest	contentTallerThanMin_growsContainer	Pass	0.231
319	GummyContainerLayoutTest	smallContent_keepsMinSize_andIsCentered	Pass	0.158
320	GummyContainerLayoutTest	wrapContentWidth_isNotZero	Pass	0.170
321	PlayItMotionTest	durationsInsideAnimationGuide	Pass	0.000
322	PlayItMotionTest	reducedMotionSnaps	Pass	0.004
323	FeedbackEffectsTest	correctPop_peaksAt108Percent	Pass	0.003
324	FeedbackEffectsTest	heartWobble_movesThenSettles	Pass	0.000
325	FeedbackEffectsTest	heartWobble_reducedMotionIsStill	Pass	0.000
326	IdleTimerTest	busyForever_givesUp	Pass	0.043
327	IdleTimerTest	busyPostponesWithoutCounting	Pass	0.049
328	IdleTimerTest	firesAfterTimeout	Pass	0.037
329	IdleTimerTest	stopCancels	Pass	0.026
330	IdleTimerTest	stopsAfterMaxPrompts	Pass	0.026
331	IdleTimerTest	touchResetsTheClock	Pass	0.030
332	IdleWiggleTest	isShort	Pass	0.000
333	IdleWiggleTest	shakesBothWays	Pass	0.001
334	IdleWiggleTest	startsAndEndsLevel	Pass	0.000
335	IdleWiggleTest	staysWithinEightDegrees	Pass	0.000
336	AssetImageTest	inSampleSize_512to120_is4	Pass	0.000
337	AssetImageTest	inSampleSize_512to200_is2	Pass	0.000
338	AssetImageTest	inSampleSize_512to512_is1	Pass	0.025
339	AssetImageTest	inSampleSize_neverZero	Pass	0.000
340	AssetImageLoadTest	backgroundLoad_newPictureReplacesTheOldOne	Pass	0.651
341	AssetImageLoadTest	missingAsset_isTransparentNotCrash	Pass	9.178

#	Test class	Test case	Result	Time (s)
342	AssetImageLoadTest	pictureIsDecodedOnFirstComposition	Pass	0.279
343	ManifestPrivacyTest	backupIsDisabled	Pass	0.032
344	ZeroEmojiPolicyTest	noEmojiInAppSources	Pass	0.503
345	PerformancePolicyTest	activityIsPortraitOnly	Pass	0.001
346	PerformancePolicyTest	noEndlessIdleAnimationOnLetterCardOrMascotHeader	Pass	0.002
347	PerformancePolicyTest	noHapticFeedbackCalls	Pass	0.219
348	PerformancePolicyTest	noIdleFloatingRequestedByCallers	Pass	0.142
349	ViewModelInitOrderTest	initBlocksComeAfterAllProperties	Pass	0.053
350	LayoutMatrixCompactTest	blendIt_tilesVisible	Pass	0.206
351	LayoutMatrixCompactTest	complete_continueVisible	Pass	0.195
352	LayoutMatrixCompactTest	complete_continueVisible_fontScale13	Skip	0.061
353	LayoutMatrixCompactTest	findIt_fontScale13	Skip	0.069
354	LayoutMatrixCompactTest	findIt_wholeGridVisible	Pass	0.221
355	LayoutMatrixCompactTest	fontScale13_mainActionVisible	Skip	0.072
356	LayoutMatrixCompactTest	hearIt_playVisible	Pass	0.115
357	LayoutMatrixCompactTest	hearIt_stepsAboveNext	Pass	0.119
358	LayoutMatrixCompactTest	lessonScaffold_bottomBarAndBodyVisible	Pass	0.093
359	LayoutMatrixCompactTest	map_buttonsAre64dp	Pass	0.248
360	LayoutMatrixCompactTest	map_currentNodeVisible	Pass	0.145
361	LayoutMatrixCompactTest	map_unitBannerTitleFits	Pass	0.118
362	LayoutMatrixCompactTest	nameprompt_letsPlayVisible	Pass	0.239
363	LayoutMatrixCompactTest	nameprompt_letsPlayVisible_fontScale13	Skip	0.061
364	LayoutMatrixCompactTest	sayIt_micVisible	Pass	0.101
365	LayoutMatrixCompactTest	sayIt_micVisible_fontScale13	Skip	0.061
366	LayoutMatrixCompactTest	sayIt_mouthCueBesideMic	Pass	0.163
367	LayoutMatrixCompactTest	splash_lilyFaceClearOfStart	Pass	0.078
368	LayoutMatrixCompactTest	splash_startVisible	Pass	0.077
369	LayoutMatrixCompactTest	splash_startVisible_fontScale13	Skip	0.095
370	LayoutMatrixCompactTest	tablet_contentWidthCapped	Pass	0.116
371	LayoutMatrixA21sTest	blendIt_tilesVisible	Pass	0.195
372	LayoutMatrixA21sTest	complete_continueVisible	Pass	0.323
373	LayoutMatrixA21sTest	complete_continueVisible_fontScale13	Pass	0.788
374	LayoutMatrixA21sTest	findIt_fontScale13	Pass	0.234
375	LayoutMatrixA21sTest	findIt_wholeGridVisible	Pass	0.387

#	Test class	Test case	Result	Time (s)
376	LayoutMatrixA21sTest	fontScale13_mainActionVisible	Pass	0.193
377	LayoutMatrixA21sTest	hearIt_playVisible	Pass	0.311
378	LayoutMatrixA21sTest	hearIt_stepsAboveNext	Pass	0.311
379	LayoutMatrixA21sTest	lessonScaffold_bottomBarAndBodyVisible	Pass	0.140
380	LayoutMatrixA21sTest	map_buttonsAre64dp	Pass	0.352
381	LayoutMatrixA21sTest	map_currentNodeVisible	Pass	0.203
382	LayoutMatrixA21sTest	map_unitBannerTitleFits	Pass	1.019
383	LayoutMatrixA21sTest	nameprompt_letsPlayVisible	Pass	0.243
384	LayoutMatrixA21sTest	nameprompt_letsPlayVisible_fontScale13	Pass	0.383
385	LayoutMatrixA21sTest	sayIt_micVisible	Pass	0.256
386	LayoutMatrixA21sTest	sayIt_micVisible_fontScale13	Pass	0.199
387	LayoutMatrixA21sTest	sayIt_mouthCueBesideMic	Pass	0.918
388	LayoutMatrixA21sTest	splash_lilyFaceClearOfStart	Pass	0.227
389	LayoutMatrixA21sTest	splash_startVisible	Pass	0.289
390	LayoutMatrixA21sTest	splash_startVisible_fontScale13	Pass	0.182
391	LayoutMatrixA21sTest	tablet_contentWidthCapped	Pass	0.117
392	LayoutMatrixPhoneTest	blendIt_tilesVisible	Pass	0.102
393	LayoutMatrixPhoneTest	complete_continueVisible	Pass	0.095
394	LayoutMatrixPhoneTest	complete_continueVisible_fontScale13	Pass	0.174
395	LayoutMatrixPhoneTest	findIt_fontScale13	Pass	0.130
396	LayoutMatrixPhoneTest	findIt_wholeGridVisible	Pass	0.585
397	LayoutMatrixPhoneTest	fontScale13_mainActionVisible	Pass	0.204
398	LayoutMatrixPhoneTest	hearIt_playVisible	Pass	0.103
399	LayoutMatrixPhoneTest	hearIt_stepsAboveNext	Pass	0.132
400	LayoutMatrixPhoneTest	lessonScaffold_bottomBarAndBodyVisible	Pass	0.059
401	LayoutMatrixPhoneTest	map_buttonsAre64dp	Pass	0.378
402	LayoutMatrixPhoneTest	map_currentNodeVisible	Pass	0.123
403	LayoutMatrixPhoneTest	map_unitBannerTitleFits	Pass	0.097
404	LayoutMatrixPhoneTest	nameprompt_letsPlayVisible	Pass	0.135
405	LayoutMatrixPhoneTest	nameprompt_letsPlayVisible_fontScale13	Pass	0.126
406	LayoutMatrixPhoneTest	sayIt_micVisible	Pass	0.179
407	LayoutMatrixPhoneTest	sayIt_micVisible_fontScale13	Pass	0.160
408	LayoutMatrixPhoneTest	sayIt_mouthCueBesideMic	Pass	0.243
409	LayoutMatrixPhoneTest	splash_lilyFaceClearOfStart	Pass	0.123

#	Test class	Test case	Result	Time (s)
410	LayoutMatrixPhoneTest	splash_startVisible	Pass	0.080
411	LayoutMatrixPhoneTest	splash_startVisible_fontScale13	Pass	0.113
412	LayoutMatrixPhoneTest	tablet_contentWidthCapped	Pass	0.231
413	LayoutMatrixTabletTest	blendIt_tilesVisible	Pass	0.167
414	LayoutMatrixTabletTest	complete_continueVisible	Pass	0.160
415	LayoutMatrixTabletTest	complete_continueVisible_fontScale13	Pass	0.148
416	LayoutMatrixTabletTest	findIt_fontScale13	Pass	0.115
417	LayoutMatrixTabletTest	findIt_wholeGridVisible	Pass	0.098
418	LayoutMatrixTabletTest	fontScale13_mainActionVisible	Pass	0.062
419	LayoutMatrixTabletTest	hearIt_playVisible	Pass	0.096
420	LayoutMatrixTabletTest	hearIt_stepsAboveNext	Pass	0.070
421	LayoutMatrixTabletTest	lessonScaffold_bottomBarAndBodyVisible	Pass	0.063
422	LayoutMatrixTabletTest	map_buttonsAre64dp	Pass	0.103
423	LayoutMatrixTabletTest	map_currentNodeVisible	Pass	0.094
424	LayoutMatrixTabletTest	map_unitBannerTitleFits	Pass	0.112
425	LayoutMatrixTabletTest	nameprompt_letsPlayVisible	Pass	0.073
426	LayoutMatrixTabletTest	nameprompt_letsPlayVisible_fontScale13	Pass	0.113
427	LayoutMatrixTabletTest	sayIt_micVisible	Pass	0.078
428	LayoutMatrixTabletTest	sayIt_micVisible_fontScale13	Pass	0.071
429	LayoutMatrixTabletTest	sayIt_mouthCueBesideMic	Pass	0.096
430	LayoutMatrixTabletTest	splash_lilyFaceClearOfStart	Pass	0.062
431	LayoutMatrixTabletTest	splash_startVisible	Pass	0.092
432	LayoutMatrixTabletTest	splash_startVisible_fontScale13	Pass	0.064
433	LayoutMatrixTabletTest	tablet_contentWidthCapped	Pass	0.055

Appendix B. How to Reproduce

```
git checkout 0320385                     # the code of debug APK H
gradlew cleanTestDebugUnitTest testDebugUnitTest
#   results: app/build/test-results/testDebugUnitTest/*.xml
gradlew recordRoborazziDebug --tests "com.playit.app.screenshot.*"
#   captures: app/build/outputs/roborazzi/*.png
python tools/docs/check_audio_release.py audio_release_check.txt
aapt dump permissions playit-debug-H-0320385.apk
python tools/docs/build_std.py --figures   # rebuilds docs/STD_v1.0.md from the logs
```